

Digital Forensic Report — Phishing-to-C2 Incident

Case ID	INC-20260927-063946 (memory), INC-20260927-072205 (network)
Affected host	WIN-5OSDUJBR958 — 10.10.12.20 (Windows Server 2019, Build 17763)
Examiner	Nguyen (nguyen)
Report version	1.1 — 2026-09-28
Classification	Confidential — lab / educational
Standards applied	NIST SP 800-86, NIST SP 800-61r3, ISO/IEC 27037, ISO/IEC 27041, ISO/IEC 27042, MITRE ATT&CK

Document control

Versi on	Date	Auth or	Change
1.0	2026-09-28	Nguy en	Initial report
1.1	2026-09-28	Nguy en	Added chain of custody, network-evidence hashes, fact/inference labelling, time-source statement, tool validation, appendix outputs

Review	Name	Date	Status
Technical review	<i>(pending)</i>		

DFIR Pipeline - Phishing-to-C2 Detection and Automated Evidence Collection

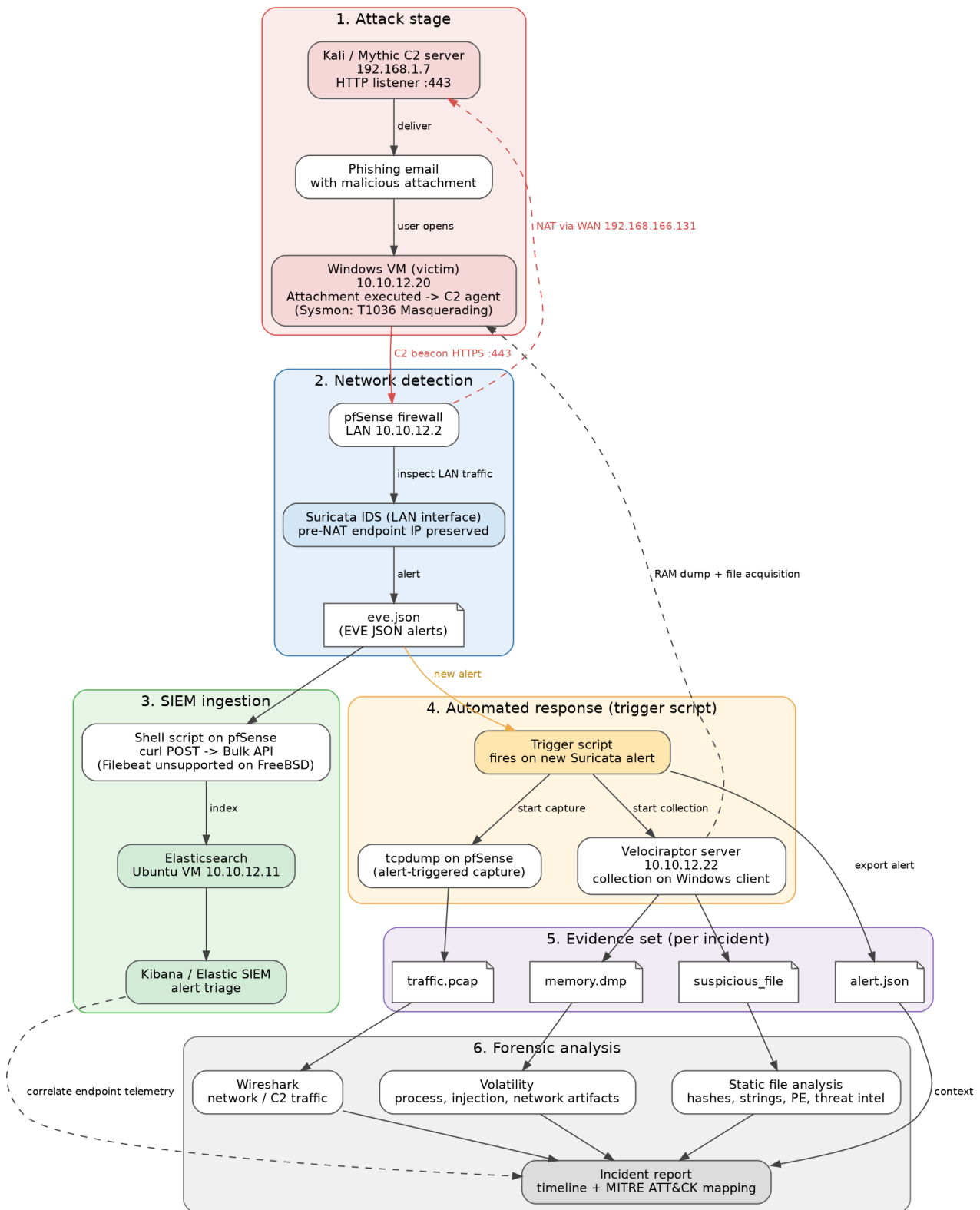


Figure 0 — Detection-and-response pipeline: Suricata alert → Elasticsearch → automated trigger → Velociraptor memory acquisition → offline analysis.

1. Executive Summary

On 2026-09-27, the automated detection pipeline flagged command-and-control (C2) traffic from the Windows endpoint **10.10.12.20** to **192.168.1.7:443**. The Suricata alert triggered automated memory acquisition through Velociraptor, and the evidence was analyzed offline with Volatility 3.

Analysis established that a **.NET executable**, **a3.exe**, launched by the interactive user from the Administrator Desktop (PID 6236), held an established session to the C2 server and spawned reconnaissance commands (`ipconfig`). Sysmon independently recorded the same binary's connection to the C2 server; the installed Sysmon configuration labelled that event with RuleName **T1036 (Masquerading)**.

A **second, independent remote-access channel** was also found: a **ScreenConnect client installed under a deceptive path** (`C:\Program Files (x86)\Microsoft Visual C++ 2015\`), connected to the external host **support.open-port-vpn.top (191.101.45.59):8041** since system boot.

Severity: High. Two concurrent remote-control channels, interactive operator activity and unbacked executable memory in the implant. Containment and re-imaging are recommended.

2. Incident Overview

2.1 Background and authorization

The investigation was performed in the examiner's SOC/DFIR lab as part of a thesis project. All systems are owned by the examiner and isolated on the VMnet1 (10.10.12.0/24) host-only network. The C2 server is a lab system operated by the examiner to generate the incident.

2.2 Investigative questions

1. Was there a genuine C2 channel from the victim endpoint?
2. Which process owned the connection, and how was it introduced?
3. Is there evidence of hands-on-keyboard activity?
4. Are there additional access channels or persistence mechanisms?
5. Which indicators of compromise (IOCs) can be extracted for detection?

2.3 Scope and limitations

- One endpoint (10.10.12.20); evidence window 2026-09-27 05:59–07:22 UTC.
- The Suricata alert covers the **network (C2) stage only**. No email or mail-gateway evidence was collected, so the delivery vector is **inferred**, not observed.
- C2 traffic is TLS-encrypted and was not decrypted; analysis relies on metadata and memory artifacts.
- The on-disk `a3.exe` was later observed as a **0-byte file** (Section 5.4); no valid on-disk hash of the live binary exists.
- The network evidence (INC-20260927-072205) and the memory image (INC-20260927-063946) come from **two different C2 sessions** (Section 4.3).
- Clock synchronisation (NTP) across lab hosts was **not verified**; see Section 2.5.

- At the time of the incident the **Velociraptor server (frontend) ran on the victim host itself** (10.10.12.20, PID 6944, ports 8000/8001/8889), not on the dedicated server 10.10.12.22. The acquired image was therefore staged on the compromised host before export. This is a custody weakness (Section 3.3) and the Velociraptor processes appear in the memory image.

2.4 Fact and inference convention (ISO/IEC 27042)

Statements in Section 5 are labelled: - **[F] Fact** — directly observed in a cited evidence item or tool output. - **[I] Inference** — the examiner's interpretation drawn from one or more facts.

2.5 Time sources

Source	Time basis in evidence	Notes
Memory image (windows.info, netscan, pstree)	UTC	Endpoint kernel clock
Sysmon	UtcTime field (UTC); Event Viewer display in local time	06:17:31 UTC = 13:17 local
Velociraptor	UTC (Z suffix)	Server clock
Trigger script / evidence directory	Server local time, converted to UTC	14:22:05 local = 07:22:05 UTC
Mythic C2 UI	Operator-host local time	Not used for timeline ordering

All hosts are in UTC+07:00. NTP synchronisation between hosts was not verified; timestamps from different hosts are ordered with an assumed tolerance of a few seconds. Where ordering depends on sub-minute differences across hosts, conclusions are stated as inferences.

2.6 Environment

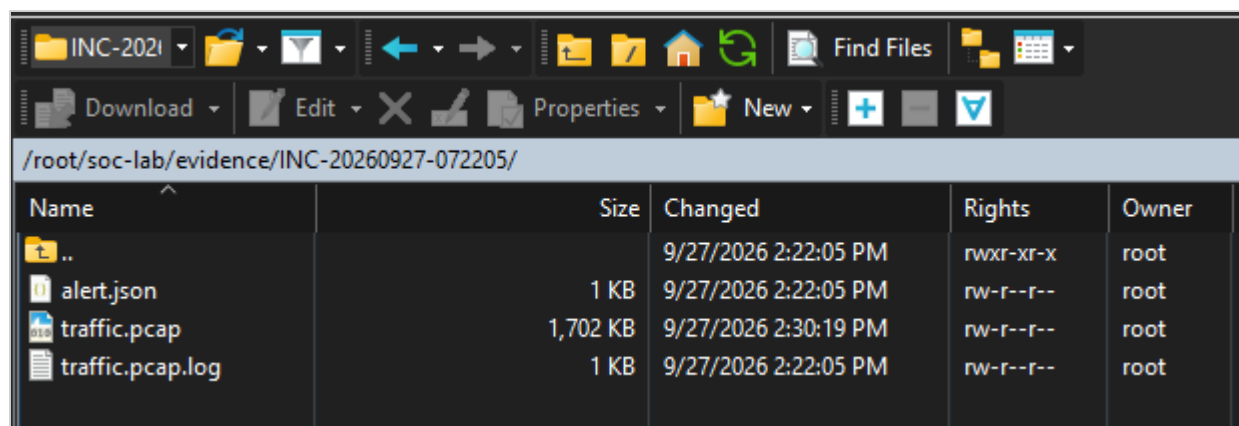
Victim 10.10.12.20, pfSense LAN 10.10.12.2, Elasticsearch/Kibana 10.10.12.11, Velociraptor 10.10.12.22, C2 192.168.1.7. See README.md for full topology.

3. Evidence Acquisition (Collection)

3.1 Evidence items

ID	Item	Source / method	Size	SHA256
E1	alert.json	Suricata EVE alert exported by trigger (INC-20260927-072205)	1 KB	bd49e274d8da4133a1ef73a7448079fdeb88c098a155b280d2a9918cbbcf7e89
E2	traffic.pcap	pfSense tcpdump, alert-triggered (INC-20260927-072205)	1,702 KB	7fa29ac565e48308d4059391b0d4f46de552b6ff4971de41bdd93618e4b90879
E3	PhysicalMemory.d d	Velociraptor Windows.Memory.Acquisition, flow F.DASBKCL6L74KI (INC-20260927-063946)	5,368,709,120 B	efa461225a760f95244952188a3c442531cad117a59598ea0701ac855108f8fd
E3-C	Velociraptor export container (ZIP) holding E3	Velociraptor download	1,415 MB compressed	d9d669078ee6df53f287d2dcac100556516a65b75111ca7ab6c8579be48e432c
E4	a3.exe on-disk state	AI-SIEM file collector, 2026-09-28	0 B	4d0ad28aea96e37a3a548c210bb7fb049ddd22a1d2aa61954ba2696f9a877b48

MD5 of E3: fb357c6474c4b42b8a528fd2d06c90de. MD5 of E4: 4cc8c5e06240f380709b690ff0de8f4b.



Name	Size	Changed	Rights	Owner
..		9/27/2026 2:22:05 PM	rxwxr-xr-x	root
alert.json	1 KB	9/27/2026 2:22:05 PM	rw-r--r--	root
traffic.pcap	1,702 KB	9/27/2026 2:30:19 PM	rw-r--r--	root
traffic.pcap.log	1 KB	9/27/2026 2:22:05 PM	rw-r--r--	root

Figure 1 — Evidence directory `/root/soc-lab/evidence/INC-20260927-072205/`: `alert.json`, `traffic.pcap`, `traffic.pcap.log`.

3.2 Acquisition method

The trigger (`suricata_velociraptor_trigger.py`) checked Elasticsearch and the Velociraptor API, matched a sev 1 C2 alert, resolved the victim IP to client **C.8f3be93e17030665**, and launched `Windows.Memory.Acquisition` (flow **F.DASBKCL6L74KI**). For the network evidence, the trigger started an alert-scoped `tcpdump` on pfSense and exported the matching EVE record.

```

(venv) root@ubuntuuser:~/soc-lab# python3 suricata_velociraptor_trigger.py --once --lookback 600
❗ KIỂM TRA KẾT NỐI...
✅ Elasticsearch OK — 3,206 alert trong 'suricata-eve*'
✅ Velociraptor API OK — 10.10.12.20:8001 (server host: WIN-50SDUJBR958)
🖥 Victim 10.10.12.20 → C.8f3be93e17030665 (WIN-50SDUJBR958)

=====
🔵 SURICATA → VELOCIRAPTOR TRIGGER STARTED
🐞 ES: http://10.10.12.11:9200/suricata-eve*
🖥 Victim: 10.10.12.20
🌱 Artifact: Windows.Memory.Acquisition
🕒 Poll: 15s | Cooldown: 30 phút
🔥 Mode: LIVE / ONE-SHOT
=====

--- Cycle #1 | 06:39:46 ---
🚨 [sev 1] C2 Mythic TLS Cert | 192.168.1.7:443 → 10.10.12.20:50083
📄 TRIGGERED Windows.Memory.Acquisition | INC-20260927-063946 | client C.8f3be93e17030665 | flow F.DASBKCL6L74KI

📊 Cycles: 1 | Alerts: 1 | Dumps triggered: 1 | Finished: 0 | Failed/Timeout: 0
⚠️ 1 flow vẫn đang chạy trên Velociraptor — xem tiếp trên GUI:
INC-20260927-063946 → flow F.DASBKCL6L74KI

```

Figure 2 — Trigger run: Elasticsearch (*suricata-eve**, 3,206 alerts) and Velociraptor API checks pass; sev 1 alert "C2 Mythic TLS Cert" (192.168.1.7:443 → 10.10.12.20:50083) fires acquisition INC-20260927-063946, flow F.DASBKCL6L74KI.

State	FlowId	Artifacts	Created	Last Active	Creator	Mb	Rows
✓	F.DASC87A9HH5K	Windows.Memory.Acquisition	2026-09-27T07:22:05.041Z	2026-09-27T07:23:37.546Z	nguyen	5120 Mb	1
✓	F.DASBKCL6L74KI	Windows.Memory.Acquisition	2026-09-27T06:39:46.203Z	2026-09-27T06:41:41.703Z	nguyen	5120 Mb	1
✓	F.DAP2N54HSTU9G	Windows.System.PowerShell	2026-09-22T07:17:40.332Z	2026-09-22T07:17:40.853Z	nguyen	7 Kb	1
✓	F.DAP025N7KJ4JU	Generic.Client.Info	2026-09-22T04:16:22.315Z	2026-09-22T04:16:23.879Z	InterrogationService	0 b	26

Overview	
Artifact Names	Windows.Memory.Acquisition
Flow ID	F.DASBKCL6L74KI
Creator	nguyen
Create Time	2026-09-27T06:39:46.203Z
Start Time	2026-09-27T06:39:46.244Z
Last Active	2026-09-27T06:41:41.703Z
Duration	115.46 seconds
State	Completed
Ops/Sec	Unlimited
CPU Limit	Unlimited
IOPS Limit	Unlimited
Timeout	600 seconds
Max Rows	1m rows
Max Mb	1000.00 Mb
Parameters	Windows.Memory.Acquisition

Results													
Artifacts with Results	Windows.Memory.Acquisition												
Total Rows	1												
Uploaded Bytes	5368789120 / 5368789120												
Files uploaded	1												
Download Results													
Available Downloads													
WIN-50SDUJBR958-C.8f3be93e17030665-F.DASBKCL6L74KI <table border="1"> <tr> <td>Uncompressed</td> <td>5120 Mb</td> </tr> <tr> <td>Compressed</td> <td>1415 Mb</td> </tr> <tr> <td>Container Files</td> <td>8</td> </tr> <tr> <td>Started</td> <td>2026-09-27T06:41:47Z</td> </tr> <tr> <td>Duration (Sec)</td> <td>90</td> </tr> <tr> <td>SHA256 Hash</td> <td>d9d669078ee6df53f287d2dcac100556516a65b75111ca7ab6c8579be48e432c</td> </tr> </table>		Uncompressed	5120 Mb	Compressed	1415 Mb	Container Files	8	Started	2026-09-27T06:41:47Z	Duration (Sec)	90	SHA256 Hash	d9d669078ee6df53f287d2dcac100556516a65b75111ca7ab6c8579be48e432c
Uncompressed	5120 Mb												
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Duration (Sec)	90												
SHA256 Hash	d9d669078ee6df53f287d2dcac100556516a65b75111ca7ab6c8579be48e432c												

Figure 3 — Velociraptor flow F.DASBKCL6L74KI: Completed, 115.46 s, 5,368,709,120 bytes uploaded, container SHA256 `d9d6...432c`.

3.3 Chain of custody (ISO/IEC 27037)

Item	Date/time (UTC)	Action	From → To	Handled by	Hash verified
E3	2026-09-27 06:39:46	Acquired from endpoint memory	Velociraptor client (PID 3176) → Velociraptor frontend on the same host, 10.10.12.20:8000	Velociraptor agent (automated, trigger)	Container hash recorded by Velociraptor
E3-C	2026-09-27 (after 06:41:41)	Exported and downloaded	Velociraptor server → analyst workstation	nguyen	Container SHA256 d9d6...432c
E3	2026-09-27	Extracted from container, image hashed before analysis	Analyst workstation	nguyen	SHA256 efa4...f8fd
E1, E2	2026-09-27 07:22:05	Written by trigger process	pfSense / SOC server → /root/soc-lab/evidence/INC-20260927-072205/	Trigger (automated)	Not hashed at creation
E1, E2	2026-09-28	Copied to analyst workstation, hashed	SOC server → C:\Users\PC\Documents\	nguyen	SHA256 in Section 3.1
E4	2026-09-28 08:53:58	Metadata and hash recorded	Endpoint → AI-SIEM	AI-SIEM collector (automated)	SHA256 recorded (0-byte file)

Custody gaps (declared): (1) The Velociraptor frontend ran on the victim host, so E3 was staged on the compromised system before export. (2) E1 and E2 were not hashed when the trigger wrote them; their first recorded hashes were taken on the analyst workstation on 2026-09-28. Integrity between creation and that point rests on access control of the SOC server (root-owned files, Figure 1). All analysis was performed on working copies; originals were not modified.

Acquisition leaves a temporary artifact on the endpoint, `C:\Windows\TEMP\tmp523269964.pmem`. This is an expected side effect of acquisition and is not attributed to the attacker.

3.4 Integrity notes

- The hash in the Velociraptor download panel (Figure 3) is the hash of the **ZIP container** (E3-C), not of the memory image (E3).
- The physical memory runs total 4,293,959,680 bytes. The image is 5 GiB because the physical address space extends to `0x140000000` and the gaps are zero-filled.

3.5 Tools and validation (ISO/IEC 27041)

Tool	Version	Purpose	Validation performed
Volatility 3	2.28.0	Memory analysis	Profile identified with windows.info (Build 17763 matches the known lab OS); process-to-connection findings cross-checked against Sysmon EID 3 and the Suricata alert
Velociraptor	0.75.x (frontend on 10.10.12.20 at time of incident)	Memory acquisition	Uploaded byte count equals expected image size (5,368,709,120 / 5,368,709,120)
Suricata	pfSense package, LAN interface	Network detection	Alert 5-tuple matches the netscan connection (192.168.1.7:443 ↔ 10.10.12.20:50083)
Wireshark	Workstation build	PCAP analysis	Conversation statistics only; no payload decryption
Sysmon	64-bit service	Endpoint telemetry	Independent source for the image path and destination
PowerShell Get-FileHash / Velociraptor	Built-in	Hashing	SHA256

windows.info output used to confirm the profile:

```
Is64Bit           True
layer_name        0 WindowsIntel32e
Major/Minor       15.17763
KeNumberProcessors 12
SystemTime        2026-09-27 06:39:47+00:00
NtProductType     NtProductServer
NtMajorVersion    10
```

Reproducibility note: on Windows, Volatility output failed with `UnicodeEncodeError` (cp1252) until `PYTHONUTF8=1` and `PYTHONIOENCODING=utf-8` were set. This affects output rendering only, not analysis results.

4. Examination

4.1 Timeline (UTC)

Time	Event	Source	Type
05:59:16	Endpoint boot	pstree (System create time)	F
05:59:22	ScreenConnect client (PID 2856) connected to 191.101.45.59:8041	netscan	F
06:17:31	Sysmon EID 3: a3.exe (PID 832) → 192.168.1.7:443, rule T1036	Sysmon	F
06:33:08	a3.exe (PID 6236) started by explorer.exe (PID 5516)	pstree	F
06:33:09	Connection 10.10.12.20:50083 → 192.168.1.7:443 by PID 6236	netscan	F
06:33:48	PID 6236 → cmd.exe (1200) → ipconfig.exe (6872)	pstree	F
06:35:45	PID 6236 → cmd.exe (1156) → ipconfig.exe (3408)	pstree	F
06:39:46	Trigger fires INC-20260927-063946; acquisition starts	trigger output, Velociraptor	F
06:39:47	Memory image system time	windows.info	F
06:41:41	Acquisition last active (115.46 s)	Velociraptor	F
07:22:05	INC-20260927-072205 evidence written (E1, E2)	evidence directory	F

The screenshot shows the Windows Event Viewer interface. The left pane displays the event log hierarchy, with 'Operational' selected. The main pane shows a list of events, with the selected event (ID 3) showing details: Network connection detected (rule: T1036, technique_names: Masquerading). The details pane shows the process path (C:\Users\Administrator\Desktop\la3.exe) and the destination IP (192.168.1.7). The right pane shows the Actions menu.

Figure 4 — Sysmon Event ID 3: RuleName `technique_id=T1036,technique_name=Masquerading`, Image `C:\Users\Administrator\Desktop\A3.exe`, `10.10.12.20:49967 → 192.168.1.7:443`, UtcTime 2026-09-27 06:17:31.247.

4.2 Correlation

The network alert and endpoint were joined on the victim IP, the C2 address `192.168.1.7:443`, and the acquisition window. The netscan row below links the Suricata alert of INC-20260927-063946 (destination port 50083) to process `a3.exe`:

Offset	Proto	LocalAddr	LPort	ForeignAddr	FPort	State	PID	Owner	Created
0x9e8a70d3eb40	TCPv4	10.10.12.20	50083	192.168.1.7	443	ESTABLISHED	6236	a3.exe	2026-09-27 06:33:09 UTC

4.3 Session mapping

Evidence from three separate C2 sessions was collected. They are not treated as one session.

Ses sion	Victim port	Time (UTC)	Evidence	Process
S1	49967	06:17:31	Sysmon EID 3	a3.exe PID 832
S2	50083	06:33:09 → acquisition 06:39:46	Suricata alert (INC-063946), netscan, E3	a3.exe PID 6236
S3	50447	~07:15-07:22	E1, E2 (INC-072205); memory acquired as flow F.DASC87A9HHH5K (07:22:05)	Memory image acquired but not analysed in this report

[I] S1-S3 share the same image path and C2 destination, which indicates repeated execution of `a3.exe` against the same C2 server. The PCAP (E2) characterises S3, not S2, and is used here as supporting evidence of C2 behaviour, not as direct confirmation of the memory findings.

5. Analysis / Findings

5.1 Finding 1 — Implant `a3.exe`

Attribute	Value
Image path	<code>C:\Users\Administrator\Desktop\A3.exe</code>
PID / PPID	6236 / 5516 (explorer.exe)
Create time	2026-09-27 06:33:08 UTC
Command line	<code>"C:\Users\Administrator\Desktop\A3.exe"</code>
Runtime	.NET Framework 4 (CLR loaded)
C2 connection	<code>10.10.12.20:50083 → 192.168.1.7:443, ESTABLISHED</code>

Process ancestry (`windows.pstree`, excerpt):

```

5516 explorer.exe
6236 a3.exe "C:\Users\Administrator\Desktop\a3.exe" 06:33:08
7096 conhost.exe
1200 cmd.exe -> 6872 ipconfig.exe 06:33:48
1156 cmd.exe -> 3408 ipconfig.exe 06:35:45

```

Loaded modules (windows.dllexport --pid 6236, excerpt):

```

clr.dll C:\Windows\Microsoft.NET\Framework64\v4.0.30319\clr.dll
clrjit.dll C:\Windows\Microsoft.NET\Framework64\v4.0.30319\clrjit.dll
winhttp.dll C:\Windows\SYSTEM32\winhttp.dll
schannel.DLL C:\Windows\system32\schannel.DLL
ncryptsslp.dll C:\Windows\system32\ncryptsslp.dll
ws2_32.dll C:\Windows\System32\ws2_32.dll

```

Unbacked executable memory (windows.malfind --pid 6236): four private regions with PAGE_EXECUTE_READWRITE protection and no file backing.

PID	Process	Start VPN	End VPN	Tag	Protection
6236	a3.exe	0x142386b0000	0x142386bffff	VadS	PAGE_EXECUTE_READWRITE
6236	a3.exe	0x14238680000	0x1423868ffff	VadS	PAGE_EXECUTE_READWRITE
6236	a3.exe	0x7ff4f1550000	0x7ff4f15effff	VadS	PAGE_EXECUTE_READWRITE
6236	a3.exe	0x7ff4f1540000	0x7ff4f154ffff	VadS	PAGE_EXECUTE_READWRITE

File object (windows.filescan): three file objects for \Users\Administrator\Desktop\a3.exe (offsets 0x9e8a705baeb0, 0x9e8a705bcad0, 0x9e8a705bd110).

Statements: - [F] a3.exe was started by explorer.exe, i.e. from the interactive user session. - [F] a3.exe owned the established connection to 192.168.1.7:443 that Suricata alerted on. - [F] a3.exe spawned cmd.exe → ipconfig.exe twice. - [F] a3.exe loaded the .NET runtime and a TLS/HTTP client stack. - [F] a3.exe contained four unbacked RWX private memory regions. - [I] a3.exe is a C2 implant, and the ipconfig executions are operator-issued commands (T1016). - [I] Unbacked RWX memory is a recognised malware indicator. Its content was not reverse-engineered in this investigation; its purpose is left undetermined.

5.2 Finding 2 — ScreenConnect under a deceptive path

Attribute	Value
Process	ScreenConnect.ClientService.exe (PID 2856), child ScreenConnect.WindowsClient.exe (PID 5808)
Path	C:\Program Files (x86)\Microsoft Visual C++ 2015\ScreenConnect.ClientService.exe
Connection	10.10.12.20:49675 → 191.101.45.59:8041, ESTABLISHED since 05:59:22 UTC
Relay parameters	h=support.open-port-vpn.top, p=8041, c=memreduct-setup; session key parameters redacted

```

0x9e8a704e58a0 TCPv4 10.10.12.20 49675 191.101.45.59 8041 ESTABLISHED 2856 ScreenConnect. 2026-09-27 05:59:22 UTC

```

Statements: - [F] A ScreenConnect client ran from a folder named for the Visual C++ runtime, not from a ScreenConnect install location. - [F] It held a session to 191.101.45.59:8041 from

boot, and its parameters name the relay `support.open-port-vpn.top`. - [I] This is remote-access-software abuse (T1219) disguised by the install path (T1036.005) and gives access that does not depend on `a3.exe`. - [F] The client was already running at boot (service create time 05:59:20 UTC), before `a3.exe` was first executed (06:17 UTC). - [F] ScreenConnect is not a component of the designed attack scenario documented in `README.md` (Mythic C2 / `a3.exe` only). - [I] Its origin is not established by the collected evidence. It predates the evidence window and may have been present in the VM base image. It is reported as an **incidental finding**: detected by the same analysis, independent of the designed scenario.

Process tree for the ScreenConnect service (`windows.pstree`, command-line key parameters redacted):

PID	PPID	Image	Create (UTC)	Path
2856	864	ScreenConnect.ClientService.exe	2026-09-27 05:59:20	C:\Program Files (x86)\Microsoft Visual C++ 2015\ScreenConnect.ClientService.exe
5808	2856	ScreenConnect.WindowsClient.exe	2026-09-27 06:04:01	C:\Program Files (x86)\Microsoft Visual C++ 2015\ScreenConnect.WindowsClient.exe

Service command line (excerpt):

```
"...\ScreenConnect.ClientService.exe" "?e=Access&y=Guest&h=support.open-port-vpn.top&p=8041&s=48aaea46-3d52-46ad-a258-ddcd1067e17c&k=[REDACTED]&v=[REDACTED]&c=memreduct-setup"
```

Parent PID 864 is `services.exe`, meaning the client runs as an installed Windows service.

5.3 Finding 3 — Benign baseline

- [F] Expected lab software was present: Elastic Agent 9.5.4 (→ 10.10.12.11:8220), Velociraptor client (PID 3176) and a Velociraptor frontend started manually from PowerShell (PID 6660/6944, `velociraptor.exe --config .\server.config.yaml frontend -v`, ports 8000/8001/8889), Sysmon64, VMware Tools, Microsoft Defender.
- [I] `svchost.exe` sessions to 4.213.25.240:443 and 150.171.28.10:443 are Microsoft service traffic and were excluded.

5.4 Finding 4 — On-disk a3.exe recorded as 0 bytes

```
--- Cycle #10 | 2026-09-28 08:54:37 | Since: 2026-09-28T08:53:53.402Z ---
Pulled 4 new log entries
Analyzing 4 log entries...
SAFE: 2 LOW: 2 MEDIUM: 0 HIGH: 0 CRITICAL: 0

DETECTION DETAILS (1 unique threats from 2 events)

Threat #1 [LOW] - Score: 38.5/100 (2 events: 2 LOW)

File: a3.exe
Path: C:\Users\Administrator\Desktop\a3.exe
Extension: exe
Size: 0 bytes
SHA256: 4d0ad28aea96e37a3a548c210bb7fb049ddd22a1d2aa61954ba2696f9a877b48
MD5: 4cc8c5e06240f380709b690ff0de8f4b
Agent: WIN-50SDUJBR958
Time: 2026-09-28T08:53:58.658Z → 2026-09-28T08:53:58.679Z

Threat: Suspicious Executable
Mô tả: File thực thi (.exe) được ML model đánh giá có đặc điểm tương tự malware dựa trên phân tích path pattern, location, và file size.

Detection Engines:
→ [ML Engine] Random Forest classifier đánh giá 77.0% khả năng là malware
→ Features kích hoạt (5 features):
→ is_executable = 1 24.6%
→ path_depth = 4 2.7%
→ file_size = 0B 2.1%
→ is_small = 1 0.5%
→ is_zero_size = 1 0.0%
→ ... +22 features = 0 (chiếm 70.1% importance còn lại)
→ [Size Engine] Executable file có size = 0 bytes. Có thể là placeholder hoặc file bị corrupt/đang được ghi

Risk Breakdown: ML=77.0×50% + IOC=0.0×20% + MITRE=0.0×30% = 38.5
Đóng góp: ML=38.5 + IOC=0.0 + MITRE=0.0 = 38.5
Phân loại: Score=38.5 → [LOW] (SAFE<20 | LOW:20-39 | MEDIUM:40-59 | HIGH:60-79 | CRITICAL≥80)

Found 1 threat(s) (2 events)! Sending to Shuffle...
Sending 1 unique alerts (deduped from 2 events)...
ALERT SENT: a3.exe | LOW | Score: 38.5 | Agent: WIN-50SDUJBR958
Sleeping 29s until next poll...
```

Figure 5 — AI-SIEM verdict for the on-disk a3.exe on 2026-09-28: 0 bytes, overall score 38.5/100 (LOW).

- [F] On 2026-09-28 the file at C:\Users\Administrator\Desktop\a3.exe was 0 bytes (E4).
- [F] The AI-SIEM rated it LOW (38.5/100); size-related features contributed most of the score.
- [I] E4's hash describes an empty file, not the executed binary, and is not a usable IOC. The binary should be recovered from E3 (windows.dmpfiles --pid 6236) and hashed.
- [I] The LOW rating is a false negative caused by the empty file.

5.5 Finding 5 – Network evidence (session S3)

Ethernet - 4	IPv4 - 6	IPv6	TC2 - 8	UDP - 2											
Address A	Port A	Address B	Port B	Packets ▼	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A	Flows
10.10.12.20	50447	192.168.1.7	443	889	161 kb	1	381	83 kb	508	78 kb	1.284410	379.8882	1738 kbits/s	1647 kbits/s	508
10.10.12.11	37222	10.10.12.20	8001	131	15 kb	0	67	7 kb	64	8 kb	0.000000	367.9454	150 bits/s	174 bits/s	50
10.10.12.11	64575	10.10.12.20	8889	68	36 kb	28	24	36 kb	44	31 kb	113.467043	15.0328	2651 bits/s	16 kbps	10
10.10.12.1	51364	10.10.12.20	8889	67	36 kb	53	23	5 kb	44	31 kb	233.461332	15.0265	2586 bits/s	16 kbps	10
10.10.12.1	53677	10.10.12.20	8889	67	36 kb	15	24	5 kb	43	31 kb	53.465272	15.0468	2717 bits/s	16 kbps	10
10.10.12.1	57316	10.10.12.20	8889	67	36 kb	65	23	5 kb	44	31 kb	293.460580	15.0392	2686 bits/s	16 kbps	10
10.10.12.1	56291	10.10.12.20	8889	66	36 kb	78	22	5 kb	44	31 kb	353.461729	15.0336	2554 bits/s	16 kbps	10
10.10.12.1	56635	10.10.12.20	8889	66	36 kb	41	22	5 kb	44	31 kb	173.469585	15.0353	2553 bits/s	16 kbps	10
10.10.12.20	49678	10.10.12.11	8220	55	6 kb	5	28	4 kb	27	2 kb	13.83638	362.9943	78 bits/s	44 bits/s	3
10.10.12.20	50457	10.10.12.11	9200	55	62 kb	10	46	61 kb	9	848 bytes	33.978231	3.1463	154 kbps	2156 bits/s	2
10.10.12.20	50478	10.10.12.11	9200	48	50 kb	45	39	49 kb	9	856 bytes	209.832888	3.0367	130 kbps	2255 bits/s	2
10.10.12.20	50465	10.10.12.11	9200	43	45 kb	22	35	44 kb	8	781 bytes	94.091385	3.0985	112 kbps	2016 bits/s	2
10.10.12.20	50472	10.10.12.11	9200	43	44 kb	34	35	43 kb	8	781 bytes	157.277940	3.0367	112 kbps	2057 bits/s	2
10.10.12.20	50486	10.10.12.11	9200	42	43 kb	58	34	42 kb	8	782 bytes	273.444072	3.0329	111 kbps	2062 bits/s	2
10.10.12.20	50494	10.10.12.11	9200	41	43 kb	71	34	43 kb	7	722 bytes	333.517429	3.0380	112 kbps	1901 bits/s	2
10.10.12.20	50453	10.10.12.11	9200	29	26 kb	3	23	26 kb	6	631 bytes	3.923210	3.0514	67 kbps	1654 bits/s	2
10.10.12.20	50466	10.10.12.11	9200	22	16 kb	23	16	16 kb	6	622 bytes	104.098418	3.0507	40 kbps	1631 bits/s	2
10.10.12.20	50454	10.10.12.11	9200	20	14 kb	6	15	14 kb	5	559 bytes	13.952774	3.0348	35 kbps	1473 bits/s	2
10.10.12.20	50463	10.10.12.11	9200	20	14 kb	20	15	14 kb	5	559 bytes	84.079824	3.0503	35 kbps	1466 bits/s	2
10.10.12.20	50477	10.10.12.11	9200	20	14 kb	44	15	13 kb	5	559 bytes	198.485600	3.0198	34 kbps	1480 bits/s	2
10.10.12.20	50482	10.10.12.11	9200	20	14 kb	54	15	14 kb	5	560 bytes	242.112659	3.0433	37 kbps	1472 bits/s	2
10.10.12.20	50484	10.10.12.11	9200	20	14 kb	56	15	13 kb	5	559 bytes	263.437412	3.0130	35 kbps	1484 bits/s	2
10.10.12.20	50492	10.10.12.11	9200	20	14 kb	69	15	14 kb	5	559 bytes	323.509436	3.0223	36 kbps	1479 bits/s	2
10.10.12.20	50499	10.10.12.11	9200	20	14 kb	81	15	13 kb	5	559 bytes	373.543222	3.0153	34 kbps	1483 bits/s	2
10.10.12.20	50455	10.10.12.11	9200	19	13 kb	7	14	13 kb	5	559 bytes	23.961488	3.0461	33 kbps	1468 bits/s	2
10.10.12.20	50458	10.10.12.11	9200	19	13 kb	11	14	13 kb	5	557 bytes	43.996773	3.0599	32 kbps	1456 bits/s	2
10.10.12.20	50459	10.10.12.11	9200	19	13 kb	16	14	13 kb	5	557 bytes	54.011106	3.0642	33 kbps	1454 bits/s	2
10.10.12.20	50460	10.10.12.11	9200	19	13 kb	17	14	13 kb	5	557 bytes	64.018689	3.0404	33 kbps	1465 bits/s	2
10.10.12.20	50462	10.10.12.11	9200	19	13 kb	19	14	13 kb	5	557 bytes	74.053540	3.0374	33 kbps	1467 bits/s	2
10.10.12.20	50467	10.10.12.11	9200	19	13 kb	29	14	12 kb	5	559 bytes	114.215493	3.0368	31 kbps	1472 bits/s	2
10.10.12.20	50468	10.10.12.11	9200	19	13 kb	30	14	12 kb	5	559 bytes	124.217344	3.0344	32 kbps	1473 bits/s	2
10.10.12.20	50469	10.10.12.11	9200	19	13 kb	31	14	13 kb	5	559 bytes	136.511607	3.0407	33 kbps	1470 bits/s	2
10.10.12.20	50471	10.10.12.11	9200	19	13 kb	33	14	13 kb	5	559 bytes	146.513172	3.0388	33 kbps	1471 bits/s	2
10.10.12.20	50474	10.10.12.11	9200	19	13 kb	36	14	12 kb	5	557 bytes	168.203737	3.0182	32 kbps	1476 bits/s	2
10.10.12.20	50475	10.10.12.11	9200	19	13 kb	42	14	13 kb	5	559 bytes	178.214226	3.0192	33 kbps	1481 bits/s	2
10.10.12.20	50476	10.10.12.11	9200	19	13 kb	43	14	12 kb	5	559 bytes	188.395106	3.1169	31 kbps	1434 bits/s	2
10.10.12.20	50480	10.10.12.11	9200	19	12 kb	47	14	12 kb	5	557 bytes	220.282674	3.0173	31 kbps	1476 bits/s	2
10.10.12.20	50481	10.10.12.11	9200	19	13 kb	48	14	13 kb	5	559 bytes	230.285279	3.0130	33 kbps	1468 bits/s	2
10.10.12.20	50483	10.10.12.11	9200	19	12 kb	55	14	12 kb	5	557 bytes	252.231227	3.0176	32 kbps	1479 bits/s	2
10.10.12.20	50483	10.10.12.11	9200	19	12 kb	60	14	12 kb	5	557 bytes	283.454626	3.0174	32 kbps	1476 bits/s	2
10.10.12.20	50489	10.10.12.11	9200	19	13 kb	66	14	13 kb	5	557 bytes	293.462419	3.0364	32 kbps	1467 bits/s	2
10.10.12.20	50490	10.10.12.11	9200	19	13 kb	67	14	13 kb	5	557 bytes	303.492073	3.0200	33 kbps	1475 bits/s	2
10.10.12.20	50491	10.10.12.11	9200	19	13 kb	68	14	13 kb	5	557 bytes	313.498609	3.0239	33 kbps	1473 bits/s	2
10.10.12.20	50495	10.10.12.11	9200	19	13 kb	72	14	12 kb	5	557 bytes	343.327550	3.0171	32 kbps	1476 bits/s	2
10.10.12.20	50497	10.10.12.11	9200	19	13 kb	79	14	12 kb	5	557 bytes	353.534159	3.0151	32 kbps	1477 bits/s	2
10.10.12.20	50498	10.10.12.11	9200	19	13 kb	80	14	12 kb	5	557 bytes	363.537767	3.0143	32 kbps	1478 bits/s	2
10.10.12.1	49727	10.10.12.20	8889	18	5 kb	75	10	3 kb	8	2 kb	353.461089	0.0333	727 kbps	513 kbps	5
10.10.12.1	51682	10.10.12.20	8889	18	5 kb	64	10	3 kb	8	2 kb	293.460089	0.0364	637 kbps	470 kbps	5
10.10.12.1	53478	10.10.12.20	8889	18	5 kb	13	10	3 kb	8	2 kb	53.464653	0.0407	583 kbps	421 kbps	5
10.10.12.1	53830	10.10.12.20	8889	18	5 kb	63	10	3 kb	8	2 kb	293.459840	0.0372	637 kbps	460 kbps	5
10.10.12.1	55112	10.10.12.20	8889	18	5 kb	62	10	3 kb	8	2 kb	293.459839	0.0346	699 kbps	494 kbps	5
10.10.12.1	57770	10.10.12.20	8889	18	5 kb	77	10	3 kb	8	2 kb	353.461352	0.0297	815 kbps	575 kbps	5
10.10.12.1	58961	10.10.12.20	8889	18	5 kb	51	10	3 kb	8	2 kb	233.460893	0.0220	1100 kbps	777 kbps	5
10.10.12.1	60033	10.10.12.20	8889	18	5 kb	50	10	3 kb	8	2 kb	233.460676	0.0236	960 kbps	723 kbps	5
10.10.12.1	62667	10.10.12.20	8889	18	5 kb	76	10	3 kb	8	2 kb	353.461351	0.0287	792 kbps	597 kbps	5
10.10.12.1	63705	10.10.12.20	8889	18	5 kb	25	10	3 kb	8	2 kb	113.464523	0.0312	761 kbps	549 kbps	5

Figure 6 — Wireshark TCP conversations for E2: the largest flow is 10.10.12.20:50447 ↔ 192.168.1.7:443, 889 packets, 161 kB, ~380 s.

- [F] E2 contains a single long-lived, low-volume TLS session between the victim and 192.168.1.7:443; the remaining flows are lab management traffic (Elasticsearch 9200, Velociraptor 8001/8889).
- [I] The pattern is consistent with C2 beaconing. It characterises session S3 (Section 4.3).

5.6 Lab ground truth (not investigative evidence)

The screenshot displays a C2 operator console interface. At the top, a table lists several callbacks from a host named WIN-50SDUJBR958. Below this, a terminal window shows the execution of the `ipconfig /all` command, displaying network configuration details for the Ethernet adapter Ethernet0.

INTERACT	IP	HOST	USER	DOMAIN
9	10.10.12.20	WIN-50SDUJBR958	Administrator	WIN-50SDUJBR958
8	10.10.12.20	WIN-50SDUJBR958	Administrator	WIN-50SDUJBR958
7	10.10.12.20	WIN-50SDUJBR958	Administrator	WIN-50SDUJBR958
6	10.10.12.20	WIN-50SDUJBR958	Administrator	WIN-50SDUJBR958
5	10.10.12.20	WIN-50SDUJBR958	Administrator	WIN-50SDUJBR958
4	10.10.12.20	WIN-50SDUJBR958	Administrator	WIN-50SDUJBR958
3	10.10.12.20	WIN-50SDUJBR958	Administrator	WIN-50SDUJBR958
2	10.10.12.20	WIN-50SDUJBR958	Administrator	WIN-50SDUJBR958
1	10.10.12.20	WIN-50SDUJBR958	Administrator	WIN-50SDUJBR958

SPLIT CALLBACK: 4 X SPLIT CALLBACK: 8 X SPLIT CALLBACK: 9 X

[Sun Sep 27 2026 02:17 AM] / T-11 / mythic_admin / C-9 / apollo
shell ipconfig/all

```
9
10 Ethernet adapter Ethernet0:
11
12 Connection-specific DNS Suffix . :
13 Description . . . . . : Intel(R) 82574L Gigabit Network
    Connection
14 Physical Address. . . . . : 00-0C-29-8C-6A-95
15 DHCP Enabled. . . . . : No
16 Autoconfiguration Enabled . . . . : Yes
17 IPv4 Address. . . . . : 10.10.12.20(Preferred)
18 Subnet Mask . . . . . : 255.255.255.0
19 Default Gateway . . . . . : 10.10.12.2
20 DNS Servers . . . . . : 10.10.12.2
21 NetBIOS over Tcpip. . . . . : Enabled
22
```

Dir: C:\Users\Administrator\Desktop

Task an agent...

Figure 7 — C2 operator console (lab ground truth): callbacks from WIN-50SDUJBR958 / 10.10.12.20 and an `ipconfig /all` task.

This screenshot comes from the examiner-operated C2 server. It would not be available in a real investigation and is not used to support any finding. It is included only to validate that the pipeline's findings match the known ground truth of the lab exercise.

6. MITRE ATT&CK Mapping

Tactic	Technique	Basis
Initial Access	T1566.001 Spearphishing Attachment	[I] inferred — no mail evidence collected
Execution	T1204.002 User Execution: Malicious File	[F] explorer.exe → a3.exe
Execution	T1059.003 Windows Command Shell	[F] a3.exe → cmd.exe → ipconfig.exe
Discovery	T1016 System Network Configuration Discovery	[F] ipconfig executions
Defense Evasion	T1036.005 Masquerading: Match Legitimate Name or Location	[F] ScreenConnect in deceptive path. Note: the T1036 RuleName on the a3.exe Sysmon event is a configuration label, not an analytic finding
Defense Evasion	T1055 / T1620 Process Injection / Reflective Loading	[I] inferred from unbacked RWX memory
Command and Control	T1071.001 Web Protocols (HTTPS)	[F] a3.exe → 192.168.1.7:443
Command and Control	T1219 Remote Access Software	[F] ScreenConnect → support.open-port-vpn.top:8041

7. Indicators of Compromise (IOCs)

Network

Type	Value	Context
IPv4 : port	192.168.1.7:443	Primary C2 (lab) — a3.exe
IPv4 : port	191.101.45.59:8041	Secondary RMM relay
Domain	support.open-port-vpn.top	ScreenConnect relay host
Victim ports	49967, 50083, 50447	Ephemeral ports of sessions S1-S3

Host

Type	Value	Context
File path	C:\Users\Administrator\Desktop\a3.exe	Implant (.NET)
File path	C:\Program Files (x86)\Microsoft Visual C++ 2015\ScreenConnect.ClientService.exe	Masqueraded RMM
Processes	a3.exe (PID 6236, PPID explorer.exe)	Implant
Memory	Unbacked RWX regions in PID 6236 (0x142386b0000, 0x14238680000, 0x7ff4f1550000, 0x7ff4f1540000)	Injection indicator

The 0-byte SHA256 (E4) is recorded in Section 3.1 for completeness but is **not** a usable IOC (Section 5.4). Prefer path, parent-process, network and memory indicators.

8. Conclusions

Against the investigative questions (Section 2.2):

1. **Yes** — [F] an established C2 session existed (`a3.exe` → 192.168.1.7:443).
2. `a3.exe`, launched interactively from the Administrator Desktop; [I] the delivery vector (phishing) is inferred, not observed.
3. **Yes** — [F] `ipconfig` reconnaissance was executed through the implant.
4. **Yes** — [F] a second, independent channel (masqueraded ScreenConnect) provided access from boot.
5. IOCs are listed in Section 7.

Confidence: **High** for findings 1–3 and the existence of the second channel (each supported by ≥2 independent sources — memory, Sysmon, network). **Medium** for the injection interpretation, which rests on memory-region attributes without binary reverse engineering.

9. Recommendations / Remediation

Containment - Isolate 10.10.12.20; preserve E3 and the disk. - Terminate/quarantine `a3.exe` (PID 6236) and the ScreenConnect service (PID 2856/5808). - Block 192.168.1.7:443, 191.101.45.59:8041 and `support.open-port-vpn.top` at pfSense.

Eradication / recovery - Re-image the endpoint; treat host Administrator credentials as compromised and rotate them. - Recover the real `a3.exe` from E3, hash it, and add the true hash to detection. - Hunt the environment for the same relay host and the deceptive ScreenConnect path.

Detection hardening - Alert on RMM binaries running outside their normal install path. - Alert on unbacked RWX allocations in .NET processes and on `explorer` → `unknown.exe` → `cmd` → `ipconfig` chains. - For encrypted C2, use JA3/JA4, TLS SNI/cert and beacon-timing analytics rather than payload signatures. - Fix the AI-SIEM 0-byte false negative: do not let size features dominate; add parent-process and network context.

Process (for a production setting) - Hash network evidence at creation, not only at analysis time (close the custody gap in Section 3.3). - Enforce and record NTP synchronisation across all sensors (Section 2.5). - Run the Velociraptor server only on the dedicated host (10.10.12.22), never on a monitored endpoint, so acquired evidence is never staged on the compromised system.

10. Appendices — Full Evidence

Every tool output and screenshot used in this report is reproduced below in full, so each finding can be checked against its source. The same content is also provided as separate files under `appendix/`. Screenshot transcriptions reproduce the text visible in the corresponding figure.

Appendix	Content	Evidence
A1	Volatility <code>windows.info</code> — image profile (<code>appendix/A1_windows_info.txt</code>)	E3
A2	Volatility <code>windows.netscan</code> — full output (<code>appendix/A2_windows_netscan.txt</code>)	E3
A3	Volatility <code>windows.pstree</code> — full output (ScreenConnect key parameters redacted) (<code>appendix/A3_windows_pstree_redacted.txt</code>)	E3
A4	Volatility <code>windows.cmdline --pid 6236</code> (<code>appendix/A4_windows_cmdline_6236.txt</code>)	E3
A5	Volatility <code>windows.dlllist --pid 6236</code> — full output (<code>appendix/A5_windows_dlllist_6236.txt</code>)	E3
A6	Volatility <code>windows.filescan</code> — matches for <code>a3.exe</code> (<code>appendix/A6_windows_filescan_a3.txt</code>)	E3
A7	Volatility <code>windows.malfind --pid 6236</code> — region table (<code>appendix/A7_windows_malfind_6236_regions.txt</code>)	E3
A8	Sysmon Event ID 3 — transcription of Figure 4 (<code>appendix/A8_sysmon_eid3.txt</code>)	Endpoint log
A9	Trigger script run — transcription of Figure 2 (<code>appendix/A9_trigger_output.txt</code>)	Trigger
A10	Velociraptor flows and acquisition metadata — transcription of Figure 3 (<code>appendix/A10_velociraptor_flows.txt</code>)	Velociraptor
A11	AI-SIEM detection output — transcription of Figure 5 (<code>appendix/A11_aisiem_detection.txt</code>)	E4
A12	Wireshark TCP conversations — transcription of Figure 6 (<code>appendix/A12_wireshark_conversations.txt</code>)	E2
A13	C2 console — lab ground truth, transcription of Figure 7 (<code>appendix/A13_mythic_ground_truth.txt</code>)	Ground truth
A14	Evidence directory listing and hashes — Figure 1 (<code>appendix/A14_evidence_directory.txt</code>)	E1, E2

A1 — Volatility windows.info — image profile

```
PS C:\Users\PC\Documents\Voltality_analysting\uploads\auto> vol -f .\PhysicalMemory.dd windows.info
Volatility 3 Framework 2.28.0
Progress: 100.00 PDB scanning finished
Variable      Value

Kernel Base   0xf8065ce00000
DTB           0x1ad000
Symbols file:///C:/Users/PC/AppData/Local/Programs/Python/Python311/Lib/site-packages/volatility3/symbols/windows/ntkrnlmp.pdb/EF9A48AFA50FF07C616585BB01919536-1.json.xz
Is64Bit       True
IsPAE         False
layer_name    0 WindowsIntel32e
memory_layer  1 FileLayer
KdVersionBlock 0xf8065d200f10
Major/Minor   15.17763
MachineType   34404
KeNumberProcessors 12
SystemTime    2026-09-27 06:39:47+00:00
NtSystemRoot  C:\Windows
NtProductType NtProductServer
NtMajorVersion 10
NtMinorVersion 0
PE MajorOperatingSystemVersion 10
PE MinorOperatingSystemVersion 0
PE Machine    34404
PE TimeDateStamp Sun Nov 10 07:20:39 2075
```

A2 — Volatility windows.netscan — full output

```
PS C:\Users\PC\Documents\Voltality_analysting\uploads\auto> vol -f .\PhysicalMemory.dd windows.netscan
Volatility 3 Framework 2.28.0
Progress: 100.00 PDB scanning finished
Offset Proto LocalAddr LocalPort ForeignAddr ForeignPort State PID
Owner Created
0x9e8a6a164b20 TCPv4 10.10.12.20 49803 4.213.25.240 443 ESTABLISHED 3064
svchost.exe 2026-09-27 06:02:20.000000 UTC
0x9e8a6a644010 TCPv4 10.10.12.20 8889 10.10.12.1 59364 CLOSED 6944 velocir
aptor.e 2026-09-27 06:40:00.000000 UTC
0x9e8a6a644460 TCPv4 10.10.12.20 49703 169.254.169.254 80 CLOSED 5664 elastic
-otel-c 2026-09-27 05:59:25.000000 UTC
0x9e8a6a654010 TCPv4 10.10.12.20 8889 10.10.12.1 49248 CLOSED 6944 velocir
aptor.e 2026-09-27 06:40:21.000000 UTC
0x9e8a6a664270 TCPv4 10.10.12.20 50134 10.10.12.11 9200 ESTABLISHED 5664
elastic-otel-c 2026-09-27 06:39:36.000000 UTC
0x9e8a6daf3050 TCPv4 0.0.0.0 135 0.0.0.0 0 LISTENING 964 svchost.exe
2026-09-27 05:59:19.000000 UTC
0x9e8a6daf3050 TCPv6 :: 135 :: 0 LISTENING 964 svchost.exe
2026-09-27 05:59:19.000000 UTC
0x9e8a6daf3590 TCPv4 0.0.0.0 49665 0.0.0.0 0 LISTENING 1288 svchost.exe
2026-09-27 05:59:19.000000 UTC
0x9e8a6daf3ad0 TCPv4 0.0.0.0 135 0.0.0.0 0 LISTENING 964 svchost.exe
2026-09-27 05:59:19.000000 UTC
0x9e8a6daf4010 TCPv4 0.0.0.0 49665 0.0.0.0 0 LISTENING 1288 svchost.exe
2026-09-27 05:59:19.000000 UTC
0x9e8a6daf4010 TCPv6 :: 49665 :: 0 LISTENING 1288 svchost.exe
2026-09-27 05:59:19.000000 UTC
0x9e8a6daf42b0 TCPv4 0.0.0.0 49666 0.0.0.0 0 LISTENING 1748 svchost.exe
2026-09-27 05:59:19.000000 UTC
0x9e8a6daf4550 TCPv4 0.0.0.0 49664 0.0.0.0 0 LISTENING 720 wininit.exe
2026-09-27 05:59:19.000000 UTC
0x9e8a6daf4550 TCPv6 :: 49664 :: 0 LISTENING 720 wininit.exe
2026-09-27 05:59:19.000000 UTC
0x9e8a6daf46a0 TCPv4 0.0.0.0 49666 0.0.0.0 0 LISTENING 1748 svchost.exe
2026-09-27 05:59:19.000000 UTC
0x9e8a6daf46a0 TCPv6 :: 49666 :: 0 LISTENING 1748 svchost.exe
2026-09-27 05:59:19.000000 UTC
0x9e8a6daf4be0 TCPv4 0.0.0.0 49664 0.0.0.0 0 LISTENING 720 wininit.exe
2026-09-27 05:59:19.000000 UTC
0x9e8a6dc66920 TCPv4 127.0.0.1 49679 127.0.0.1 49741 ESTABLISHED 5664
elastic-otel-c 2026-09-27 05:59:26.000000 UTC
0x9e8a6f4c6bb0 TCPv4 10.10.12.20 50142 10.10.12.11 9200 ESTABLISHED 5664
elastic-otel-c 2026-09-27 06:40:45.000000 UTC
0x9e8a6f6a5b80 TCPv4 10.10.12.20 8889 10.10.12.1 52225 CLOSED 6944 velocir
aptor.e 2026-09-27 06:40:00.000000 UTC
0x9e8a6f6f66e0 TCPv4 0.0.0.0 49667 0.0.0.0 0 LISTENING 2736 spoolsv.exe
2026-09-27 05:59:20.000000 UTC
0x9e8a6f6f6830 TCPv4 0.0.0.0 49667 0.0.0.0 0 LISTENING 2736 spoolsv.exe
2026-09-27 05:59:20.000000 UTC
0x9e8a6f6f6830 TCPv6 :: 49667 :: 0 LISTENING 2736 spoolsv.exe
2026-09-27 05:59:20.000000 UTC
0x9e8a6f6f72b0 TCPv4 0.0.0.0 49668 0.0.0.0 0 LISTENING 864 services.exe
2026-09-27 05:59:20.000000 UTC
0x9e8a6f6f72b0 TCPv6 :: 49668 :: 0 LISTENING 864 services.exe
2026-09-27 05:59:20.000000 UTC
0x9e8a6f6f77f0 TCPv4 0.0.0.0 49668 0.0.0.0 0 LISTENING 864 services.exe
```

2026-09-27 05:59:20.000000 UTC	0x9e8a6f7a90	TCPv4	10.10.12.20	139	0.0.0.0 0	LISTENING	4	System	
2026-09-27 05:59:20.000000 UTC	0x9e8a6f7be0	TCPv4	0.0.0.0 445	0.0.0.0 0		LISTENING	4	System	2026-09
-27 05:59:20.000000 UTC	0x9e8a6f7be0	TCPv6	:: 445	:: 0		LISTENING	4	System	2026-09
-27 05:59:20.000000 UTC	0x9e8a6f7d30	TCPv4	0.0.0.0 47001	0.0.0.0 0		LISTENING	4	System	2026-09
-27 05:59:20.000000 UTC	0x9e8a6f7d30	TCPv6	:: 47001	:: 0		LISTENING	4	System	2026-09
-27 05:59:20.000000 UTC	0x9e8a6f7e80	TCPv4	0.0.0.0 5985	0.0.0.0 0		LISTENING	4	System	2026-09
-27 05:59:20.000000 UTC	0x9e8a6f7e80	TCPv6	:: 5985	:: 0		LISTENING	4	System	2026-09
-27 05:59:20.000000 UTC	0x9e8a6f8cfba0	TCPv4	10.10.12.20	50141	10.10.12.11	9200	CLOSED	5664	elastic
-otel-c 2026-09-27 06:40:37.000000 UTC	0x9e8a6f978cb0	UDPv4	0.0.0.0 0	*	0	6236	a3.exe		2026-09-27 06:3
3:09.000000 UTC	0x9e8a6f9796c0	UDPv4	0.0.0.0 0	*	0	6236	a3.exe		2026-09-27 06:3
3:09.000000 UTC	0x9e8a6f9796c0	UDPv6	:: 0	*	0	6236	a3.exe		2026-09-27 06:3
3:09.000000 UTC	0x9e8a6f97a0d0	UDPv4	0.0.0.0 0	*	0	6236	a3.exe		2026-09-27 06:3
3:09.000000 UTC	0x9e8a6f97a690	UDPv4	0.0.0.0 0	*	0	6236	a3.exe		2026-09-27 06:3
3:09.000000 UTC	0x9e8a6f97a690	UDPv6	:: 0	*	0	6236	a3.exe		2026-09-27 06:3
3:09.000000 UTC	0x9e8a6f983650	UDPv4	0.0.0.0 0	*	0	3040	svchost.exe		2026-09
-27 05:59:20.000000 UTC	0x9e8a6f98c4a0	UDPv4	0.0.0.0 0	*	0	3040	svchost.exe		2026-09
-27 05:59:20.000000 UTC	0x9e8a6f98c4a0	UDPv6	:: 0	*	0	3040	svchost.exe		2026-09
-27 05:59:20.000000 UTC	0x9e8a6fb200a0	UDPv4	0.0.0.0 0	*	0	2916	svchost.exe		2026-09
-27 05:59:20.000000 UTC	0x9e8a6fb200a0	UDPv6	:: 0	*	0	2916	svchost.exe		2026-09
-27 05:59:20.000000 UTC	0x9e8a6fb20940	UDPv4	0.0.0.0 0	*	0	2916	svchost.exe		2026-09
-27 05:59:20.000000 UTC	0x9e8a6fb21bf0	UDPv4	0.0.0.0 0	*	0	2916	svchost.exe		2026-09
-27 05:59:20.000000 UTC	0x9e8a6fb22040	UDPv4	0.0.0.0 0	*	0	2916	svchost.exe		2026-09
-27 05:59:20.000000 UTC	0x9e8a6fb22490	UDPv4	0.0.0.0 0	*	0	2916	svchost.exe		2026-09
-27 05:59:20.000000 UTC	0x9e8a6fb22490	UDPv6	:: 0	*	0	2916	svchost.exe		2026-09
-27 05:59:20.000000 UTC	0x9e8a6fb24430	UDPv4	0.0.0.0 0	*	0	2916	svchost.exe		2026-09
-27 05:59:20.000000 UTC	0x9e8a6fb24430	UDPv6	:: 0	*	0	2916	svchost.exe		2026-09
-27 05:59:20.000000 UTC	0x9e8a6fb256e0	UDPv4	0.0.0.0 0	*	0	1904	svchost.exe		2026-09
-27 05:59:20.000000 UTC	0x9e8a6fb256e0	UDPv6	:: 0	*	0	1904	svchost.exe		2026-09
-27 05:59:20.000000 UTC	0x9e8a6fb26260	UDPv4	0.0.0.0 0	*	0	1904	svchost.exe		2026-09
-27 05:59:20.000000 UTC	0x9e8a6fb26c70	UDPv4	0.0.0.0 0	*	0	4	System		2026-09-27 05:5
9:20.000000 UTC									

0x9e8a6fb28aa0	UDPv4	0.0.0.0	0	*	0	3040	svchost.exe	2026-09-27 05:59:20.000000 UTC
0x9e8a6fb28aa0	UDPv6	::	0	*	0	3040	svchost.exe	2026-09-27 05:59:20.000000 UTC
0x9e8a6fb2abb0	UDPv4	0.0.0.0	0	*	0	4	System	2026-09-27 05:59:20.000000 UTC
0x9e8a6fb2ba10	UDPv4	0.0.0.0	0	*	0	3040	svchost.exe	2026-09-27 05:59:20.000000 UTC
0x9e8a6fb307b0	UDPv4	0.0.0.0	0	*	0	1904	svchost.exe	2026-09-27 05:59:20.000000 UTC
0x9e8a6fb342a0	UDPv4	0.0.0.0	0	*	0	3548	svchost.exe	2026-09-27 05:59:20.000000 UTC
0x9e8a6fb34410	UDPv4	0.0.0.0	0	*	0	3608	svchost.exe	2026-09-27 05:59:20.000000 UTC
0x9e8a6fb363b0	UDPv4	0.0.0.0	0	*	0	3548	svchost.exe	2026-09-27 05:59:20.000000 UTC
0x9e8a6fb363b0	UDPv6	::	0	*	0	3548	svchost.exe	2026-09-27 05:59:20.000000 UTC
0x9e8a6fdf7d70	TCPv4	0.0.0.0	5357	0.0.0.0	0	LISTENING	4	System 2026-09-27 05:59:21.000000 UTC
0x9e8a6fdf7d70	TCPv6	::	5357	::	0	LISTENING	4	System 2026-09-27 05:59:21.000000 UTC
0x9e8a6fdf7ec0	TCPv4	127.0.0.1	6791	0.0.0.0	0	LISTENING	3168	elastic-agent. 2026-09-27 05:59:22.000000 UTC
0x9e8a6fdf8be0	TCPv4	127.0.0.1	6789	0.0.0.0	0	LISTENING	3168	elastic-agent. 2026-09-27 05:59:22.000000 UTC
0x9e8a70041b30	TCPv4	10.10.12.20	8000	10.10.12.20	49867	ESTABLISHED	6944	velociraptor.e 2026-09-27 06:05:40.000000 UTC
0x9e8a700648b0	TCPv4	127.0.0.1	8001	127.0.0.1	49853	ESTABLISHED	6944	velociraptor.e 2026-09-27 06:04:49.000000 UTC
0x9e8a704e58a0	TCPv4	10.10.12.20	49675	191.101.45.59	8041	ESTABLISHED	2856	ScreenConnect. 2026-09-27 05:59:22.000000 UTC
0x9e8a704e88a0	TCPv4	10.10.12.20	49678	10.10.12.11	8220	ESTABLISHED	3168	elastic-agent. 2026-09-27 05:59:22.000000 UTC
0x9e8a70516460	TCPv4	127.0.0.1	49770	127.0.0.1	6791	ESTABLISHED	5664	elastic-otel-c 2026-09-27 05:59:32.000000 UTC
0x9e8a705592a0	UDPv4	0.0.0.0	0	*	0	5104	svchost.exe	2026-09-27 05:59:21.000000 UTC
0x9e8a7055ab10	UDPv4	0.0.0.0	0	*	0	5104	svchost.exe	2026-09-27 05:59:21.000000 UTC
0x9e8a7055adf0	UDPv4	0.0.0.0	0	*	0	5104	svchost.exe	2026-09-27 05:59:21.000000 UTC
0x9e8a7055adf0	UDPv6	::	0	*	0	5104	svchost.exe	2026-09-27 05:59:21.000000 UTC
0x9e8a7055b0d0	UDPv4	0.0.0.0	0	*	0	5104	svchost.exe	2026-09-27 05:59:21.000000 UTC
0x9e8a7055b0d0	UDPv6	::	0	*	0	5104	svchost.exe	2026-09-27 05:59:21.000000 UTC
0x9e8a7055b3b0	UDPv4	0.0.0.0	0	*	0	5104	svchost.exe	2026-09-27 05:59:21.000000 UTC
0x9e8a7055b3b0	UDPv6	::	0	*	0	5104	svchost.exe	2026-09-27 05:59:21.000000 UTC
0x9e8a7055b690	UDPv4	0.0.0.0	0	*	0	5104	svchost.exe	2026-09-27 05:59:21.000000 UTC
0x9e8a70599270	TCPv4	10.10.12.20	49867	10.10.12.20	8000	ESTABLISHED	3176	Velociraptor.e 2026-09-27 06:05:40.000000 UTC
0x9e8a707f5050	TCPv4	0.0.0.0	49767	0.0.0.0	0	LISTENING	900	lsass.exe 2026-09-27 05:59:28.000000 UTC
0x9e8a707f52f0	TCPv4	127.0.0.1	49679	0.0.0.0	0	LISTENING	5664	elastic-otel-c 2026-09-27 05:59:26.000000 UTC
0x9e8a707f5590	TCPv4	127.0.0.1	49687	0.0.0.0	0	LISTENING	5664	elastic

```

-otel-c 2026-09-27 05:59:25.000000 UTC
0x9e8a707f5d70 TCPv4 0.0.0.0 8889 0.0.0.0 0 LISTENING 6944 velociraptor.e
2026-09-27 06:04:49.000000 UTC
0x9e8a707f5d70 TCPv6 :: 8889 :: 0 LISTENING 6944 velociraptor.e
2026-09-27 06:04:49.000000 UTC
0x9e8a707f6160 TCPv4 0.0.0.0 49767 0.0.0.0 0 LISTENING 900 lsass.exe
2026-09-27 05:59:28.000000 UTC
0x9e8a707f6160 TCPv6 :: 49767 :: 0 LISTENING 900 lsass.exe
2026-09-27 05:59:28.000000 UTC
0x9e8a707f6400 TCPv4 0.0.0.0 8001 0.0.0.0 0 LISTENING 6944 velociraptor.e
2026-09-27 06:04:49.000000 UTC
0x9e8a707f6400 TCPv6 :: 8001 :: 0 LISTENING 6944 velociraptor.e
2026-09-27 06:04:49.000000 UTC
0x9e8a707f67f0 TCPv4 127.0.0.1 8003 0.0.0.0 0 LISTENING 6944 velocir
aport.e 2026-09-27 06:04:49.000000 UTC
0x9e8a707f6be0 TCPv4 0.0.0.0 8000 0.0.0.0 0 LISTENING 6944 velociraptor.e
2026-09-27 06:04:49.000000 UTC
0x9e8a707f6be0 TCPv6 :: 8000 :: 0 LISTENING 6944 velociraptor.e
2026-09-27 06:04:49.000000 UTC
0x9e8a70b0fb80 TCPv4 10.10.12.20 49869 10.10.12.20 8000 ESTABLISHED 3176
Velociraptor.e 2026-09-27 06:05:46.000000 UTC
0x9e8a70b268a0 TCPv4 127.0.0.1 49853 127.0.0.1 8001 ESTABLISHED 6944
velociraptor.e 2026-09-27 06:04:49.000000 UTC
0x9e8a70b2e010 TCPv4 10.10.12.20 49764 169.254.169.254 80 CLOSED 5664 elastic
-otel-c 2026-09-27 05:59:28.000000 UTC
0x9e8a70ba6460 TCPv4 10.10.12.20 49694 169.254.169.254 80 CLOSED 5664 elastic
-otel-c 2026-09-27 05:59:25.000000 UTC
0x9e8a70ba8010 TCPv4 127.0.0.1 6791 127.0.0.1 49770 ESTABLISHED 3168
elastic-agent. 2026-09-27 05:59:32.000000 UTC
0x9e8a70bc3010 TCPv4 10.10.12.20 49747 169.254.169.254 80 CLOSED 5664 elastic
-otel-c 2026-09-27 05:59:26.000000 UTC
0x9e8a70becba0 TCPv4 10.10.12.20 8889 10.10.12.1 50160 CLOSED 6944 velocir
aport.e 2026-09-27 06:38:58.000000 UTC
0x9e8a70c7e460 TCPv4 10.10.12.20 49732 169.254.169.254 80 CLOSED 5664 elastic
-otel-c 2026-09-27 05:59:25.000000 UTC
0x9e8a70c82460 TCPv4 127.0.0.1 49741 127.0.0.1 49679 ESTABLISHED 3168
elastic-agent. 2026-09-27 05:59:26.000000 UTC
0x9e8a70d3eb40 TCPv4 10.10.12.20 50083 192.168.1.7 443 ESTABLISHED 6236
a3.exe 2026-09-27 06:33:09.000000 UTC
0x9e8a70e7d920 TCPv4 10.10.12.20 8000 10.10.12.20 49869 ESTABLISHED 6944
velociraptor.e 2026-09-27 06:05:46.000000 UTC
0x9e8a72b9d920 TCPv4 10.10.12.20 49840 150.171.28.10 443 CLOSED - -
2026-09-27 06:04:17.000000 UTC
0x9e8a72b9e920 TCPv4 10.10.12.20 49841 150.171.28.10 443 CLOSED - -
2026-09-27 06:04:18.000000 UTC
0x9e8a73e35ba0 TCPv4 10.10.12.20 8889 10.10.12.1 54534 CLOSED 6944 velocir
aport.e 2026-09-27 06:40:21.000000 UTC
0xc480001bdd70 TCPv4 0.0.0.0 5357 0.0.0.0 0 LISTENING 4 System 2026-09
-27 05:59:21.000000 UTC
0xc480001bdd70 TCPv6 :: 5357 :: 0 LISTENING 4 System 2026-09
-27 05:59:21.000000 UTC
0xc480001bdec0 TCPv4 127.0.0.1 6791 0.0.0.0 0 LISTENING 3168 elastic
-agent. 2026-09-27 05:59:22.000000 UTC
0xc480001bebe0 TCPv4 127.0.0.1 6789 0.0.0.0 0 LISTENING 3168 elastic
-agent. 2026-09-27 05:59:22.000000 UTC
0xf8065d7e0460 TCPv4 10.10.12.20 49732 169.254.169.254 80 CLOSED 5664 elastic
-otel-c 2026-09-27 05:59:25.000000 UTC
0xf8065d7e4460 TCPv4 127.0.0.1 49741 127.0.0.1 49679 ESTABLISHED 3168
elastic-agent. 2026-09-27 05:59:26.000000 UTC
0xf8065d866050 TCPv4 0.0.0.0 49767 0.0.0.0 0 LISTENING 900 lsass.exe
2026-09-27 05:59:28.000000 UTC

```

0xf8065d8662f0	TCPv4	127.0.0.1	49679	0.0.0.0	0	LISTENING	5664	elastic
-otel-c 2026-09-27 05:59:26.000000 UTC								
0xf8065d866590	TCPv4	127.0.0.1	49687	0.0.0.0	0	LISTENING	5664	elastic
-otel-c 2026-09-27 05:59:25.000000 UTC								
0xf8065d866d70	TCPv4	0.0.0.0	8889	0.0.0.0	0	LISTENING	6944	velociraptor.e
2026-09-27 06:04:49.000000 UTC								
0xf8065d866d70	TCPv6	::	8889	::	0	LISTENING	6944	velociraptor.e
2026-09-27 06:04:49.000000 UTC								
0xf8065d867160	TCPv4	0.0.0.0	49767	0.0.0.0	0	LISTENING	900	lsass.exe
2026-09-27 05:59:28.000000 UTC								
0xf8065d867160	TCPv6	::	49767	::	0	LISTENING	900	lsass.exe
2026-09-27 05:59:28.000000 UTC								
0xf8065d867400	TCPv4	0.0.0.0	8001	0.0.0.0	0	LISTENING	6944	velociraptor.e
2026-09-27 06:04:49.000000 UTC								
0xf8065d867400	TCPv6	::	8001	::	0	LISTENING	6944	velociraptor.e
2026-09-27 06:04:49.000000 UTC								
0xf8065d8677f0	TCPv4	127.0.0.1	8003	0.0.0.0	0	LISTENING	6944	velociraptor.e
2026-09-27 06:04:49.000000 UTC								
0xf8065d867be0	TCPv4	0.0.0.0	8000	0.0.0.0	0	LISTENING	6944	velociraptor.e
2026-09-27 06:04:49.000000 UTC								
0xf8065d867be0	TCPv6	::	8000	::	0	LISTENING	6944	velociraptor.e
2026-09-27 06:04:49.000000 UTC								
0xf8065d901ba0	TCPv4	10.10.12.20	8889	10.10.12.1	50160	CLOSED	6944	velociraptor.e
2026-09-27 06:38:58.000000 UTC								

A3 — Volatility `windows.pstree` — full output (ScreenConnect key parameters)

redacted)

```
PS C:\Users\PC\Documents\Voltality_analysting\uploads\auto> vol -f .\PhysicalMemory.dd windows.
pstree
Volatility 3 Framework 2.28.0
Progress: 100.00 PDB scanning finished
PID PPID ImageFileName Offset(V) Threads Handles SessionId Wow64 CreateT
ime ExitTime Audit Cmd Path
4 0 System 0x9e8a6a07f040 288 - N/A False 2026-09-27 05:59:16.000
000 UTC N/A - - -
* 168 4 Registry 0x9e8a6a0c8080 4 - N/A False 2026-09-27 05:5
9:11.000000 UTC N/A Registry - -
* 488 4 smss.exe 0x9e8a6d31d200 2 - N/A False 2026-09-27 05:5
9:16.000000 UTC N/A \Device\HarddiskVolume4\Windows\System32\smss.exe \SystemRoot\Sy
stem32\smss.exe \SystemRoot\System32\smss.exe
612 596 csrss.exe 0x9e8a6dccb140 15 - 0 False 2026-09-27 05:5
9:18.000000 UTC N/A \Device\HarddiskVolume4\Windows\System32\csrss.exe %SystemRoot%\s
ystem32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,20480,768 Windows=0n SubSystemTyp
e=Windows ServerDll=basesrv,1 ServerDll=winsrv:UserServerDllInitialization,3 ServerDll=sxssrv,4
ProfileControl=0ff MaxRequestThreads=16 C:\Windows\system32\csrss.exe
720 596 wininit.exe 0x9e8a6d9d70c0 1 - 0 False 2026-09-27 05:5
9:18.000000 UTC N/A \Device\HarddiskVolume4\Windows\System32\wininit.exe wininit.exe
C:\Windows\system32\wininit.exe
* 864 720 services.exe 0x9e8a6ebef6c0 5 - 0 False 2026-09-27 05:5
9:18.000000 UTC N/A \Device\HarddiskVolume4\Windows\System32\services.exe C:\Windows\sys
tem32\services.exeC:\Windows\system32\services.exe
** 1536 864 svchost.exe 0x9e8a6f4f02c0 5 - 0 False 2026-09-27 05:5
9:19.000000 UTC N/A \Device\HarddiskVolume4\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k LocalServiceNetworkRestricted -p -s Dhcp C:\Windows\system32\svchost.
exe
** 2944 864 vm3dservice.ex 0x9e8a6f99d0c0 3 - 0 False 2026-09-27 05:5
9:20.000000 UTC N/A \Device\HarddiskVolume4\Program Files\VMware\VMware Tools\vm3dservice.
exe "C:\Program Files\VMware\VMware Tools\vm3dservice.exe" C:\Program Files\VMware\VMwa
re Tools\vm3dservice.exe
*** 3336 2944 vm3dservice.ex 0x9e8a6fa252c0 4 - 1 False 2026-09
-27 05:59:20.000000 UTC N/A \Device\HarddiskVolume4\Program Files\VMware\VMware Tools\vm3d
service.exe "C:\Program Files\VMware\VMware Tools\vm3dservice.exe" -n C:\Program F
iles\VMware\VMware Tools\vm3dservice.exe
** 5248 864 msdtc.exe 0x9e8a705ec280 9 - 0 False 2026-09-27 05:5
9:21.000000 UTC N/A \Device\HarddiskVolume4\Windows\System32\msdtc.exe C:\Windows\Sys
tem32\msdtc.exe C:\Windows\System32\msdtc.exe
** 6016 864 svchost.exe 0x9e8a6ec2c340 3 - 0 False 2026-09-27 06:0
4:01.000000 UTC N/A \Device\HarddiskVolume4\Windows\System32\svchost.exe C:\Windows\Sys
tem32\svchost.exe -k LocalSystemNetworkRestricted -p -s TabletInputService C:\Windows\System32
\svchost.exe
*** 5440 6016 ctfmon.exe 0x9e8a6fc93600 11 - 1 False 2026-09
-27 06:04:01.000000 UTC N/A \Device\HarddiskVolume4\Windows\System32\ctfmon.exe "ctfmo
n.exe" C:\Windows\system32\ctfmon.exe
** 2052 864 svchost.exe 0x9e8a6a05e600 4 - 0 False 2026-09-27 05:5
9:19.000000 UTC N/A \Device\HarddiskVolume4\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k LocalServiceNetworkRestricted -p -s WinHttpAutoProxySvc C:\Windows\s
ystem32\svchost.exe
** 1032 864 svchost.exe 0x9e8a6dd99380 4 - 0 False 2026-09-27 05:5
9:19.000000 UTC N/A \Device\HarddiskVolume4\Windows\System32\svchost.exe C:\Windows\sys
tem32\svchost.exe -k DcomLaunch -p -s LSM C:\Windows\system32\svchost.exe
** 1288 864 svchost.exe 0x9e8a6f40d2c0 8 - 0 False 2026-09-27 05:5
9:19.000000 UTC N/A \Device\HarddiskVolume4\Windows\System32\svchost.exe C:\Windows\Sys
tem32\svchost.exe -k LocalServiceNetworkRestricted -p -s EventLog C:\Windows\System32\svchost.
exe
```

```

** 1544 864      svchost.exe      0x9e8a6f514540 3      -      0      False  2026-09-27 05:5
9:19.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k netsvcs -p -s gpsvc      C:\Windows\system32\svchost.exe
** 2952 864      VGAuthService.  0x9e8a6f9340c0 2      -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Program Files\VMware\VMware Tools\VMware VGAut
h\VGAuthService.exe"C:\Program Files\VMware\VMware Tools\VMware VGAuth\VGAuthService.exe" C:\P
rogram Files\VMware\VMware Tools\VMware VGAuth\VGAuthService.exe
** 140 864      svchost.exe      0x9e8a6ec36240 1      -      0      False  2026-09-27 05:5
9:18.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k DcomLaunch -p -s PlugPlay      C:\Windows\system32\svchost.exe
** 1296 864      svchost.exe      0x9e8a6f43e2c0 2      -      0      False  2026-09-27 05:5
9:19.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k LocalServiceNetworkRestricted -p -s TimeBrokerSvc      C:\Windows\system32
\svchost.exe
** 2960 864      MsMpEng.exe      0x9e8a6f960300 8      -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\ProgramData\Microsoft\Windows Defender\Platfor
m\4.18.26080.4-0\MsMpEng.exe      "C:\ProgramData\Microsoft\Windows Defender\platform\4.18.260
80.4-0\MsMpEng.exe" C:\ProgramData\Microsoft\Windows Defender\platform\4.18.26080.4-0\MsMpEng.e
xe
*** 5472      2960      DefenderSessio 0x9e8a6a631080 1      -      1      False  2026-09
-27 06:04:01.000000 UTC  N/A      \Device\HarddiskVolume4\ProgramData\Microsoft\Windows Defender
\Platform\4.18.26080.4-0\DefenderSessionHelper.exe "C:\ProgramData\Microsoft\Windows Defender\p
latform\4.18.26080.4-0\DefenderSessionHelper.exe" C:\ProgramData\Microsoft\Windows Defender\p
latform\4.18.26080.4-0\DefenderSessionHelper.exe
** 1168 864      svchost.exe      0x9e8a6fb5d080 3      -      0      False  2026-09-27 06:0
1:26.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k LocalService -p -s CDPSvc      C:\Windows\system32\svchost.exe
** 2324 864      svchost.exe      0x9e8a6f7162c0 5      -      0      False  2026-09-27 05:5
9:19.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\Sys
tem32\svchost.exe -k NetworkService -p -s LanmanWorkstation      C:\Windows\System32\svchost.
exe
** 1304 864      svchost.exe      0x9e8a6f4652c0 2      -      0      False  2026-09-27 05:5
9:19.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k LocalServiceNoNetwork -p      C:\Windows\system32\svchost.exe
** 2968 864      MpDefenderCore 0x9e8a6f962380 8      -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\ProgramData\Microsoft\Windows Defender\Platfor
m\4.18.26080.4-0\MpDefenderCoreService.exe "C:\ProgramData\Microsoft\Windows Defender\platform
\4.18.26080.4-0\MpDefenderCoreService.exe" C:\ProgramData\Microsoft\Windows Defender\platform
\4.18.26080.4-0\MpDefenderCoreService.exe
** 3608 864      svchost.exe      0x9e8a6fbf0240 5      -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\Sys
tem32\svchost.exe -k NetSvcs -p -s iphlpsvc      C:\Windows\System32\svchost.exe
** 2976 864      Sysmon64.exe     0x9e8a6f968600 17      -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\Sysmon64.exe      C:\Windows\Sysmon64.ex
e C:\Windows\Sysmon64.exe
** 2852 864      svchost.exe      0x9e8a6a665080 6      -      0      False  2026-09-27 06:0
1:21.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\Sys
tem32\svchost.exe -k LocalServiceNoNetwork -p -s DPS      C:\Windows\System32\svchost.exe
** 2088 864      svchost.exe      0x9e8a6f721240 2      -      0      False  2026-09-27 05:5
9:19.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\Sys
tem32\svchost.exe -k netsvcs -p -s ShellHWDetection      C:\Windows\System32\svchost.exe
** 2856 864      ScreenConnect.  0x9e8a6f937680 14      -      0      True   2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Program Files (x86)\Microsoft Visual C++ 2015
\ScreenConnect.ClientService.exe "C:\Program Files (x86)\Microsoft Visual C++ 2015\ScreenCon
nect.ClientService.exe" "?e=Access&y=Guest&h=support.open-port-vpn.top&p=8041&s=48aaea46-3d52-4
6ad-a258-ddcd1067e17c&k=[REDACTED]&v=[REDACTED]&c=memreduct-setup" C:\Program Files (x8
6)\Microsoft Visual C++ 2015\ScreenConnect.ClientService.exe
*** 5808      2856      ScreenConnect.  0x9e8a7006d080 7      -      1      False  2026-09
-27 06:04:01.000000 UTC  N/A      \Device\HarddiskVolume4\Program Files (x86)\Microsoft Visual C
++ 2015\ScreenConnect.WindowsClient.exe "C:\Program Files (x86)\Microsoft Visual C++ 2015\Sc
reenConnect.WindowsClient.exe" "RunRole" "f05a3326-a87e-4e8b-b27f-015285ef97f8" "User"

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C:\Program Files (x86)\Microsoft Visual C++ 2015\ScreenConnect.WindowsClient.exe
** 3880 864      svchost.exe      0x9e8a6fd12600 3      -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\Sys
tem32\svchost.exe -k LocalServiceNetworkRestricted -p -s lmhosts      C:\Windows\System32\svchost.
exe
** 1708 864      svchost.exe      0x9e8a6f5b92c0 5      -      0      False  2026-09-27 05:5
9:19.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\Sys
tem32\svchost.exe -k NetworkService -p -s NlaSvc      C:\Windows\System32\svchost.exe
** 2736 864      spoolsv.exe      0x9e8a6f8a7200 7      -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\spoolsv.exe      C:\Windows\Sys
tem32\spoolsv.exe C:\Windows\System32\spoolsv.exe
** 3760 864      svchost.exe      0x9e8a6fc62200 12     -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\Sys
tem32\svchost.exe -k netsvcs      C:\Windows\System32\svchost.exe
** 944 864       svchost.exe      0x9e8a70b27680 2      -      1      False  2026-09-27 06:0
4:01.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k UnistackSvcGroup -s WpnUserService      C:\Windows\system32\svchost.exe
** 1204 864      svchost.exe      0x9e8a6edf5240 1      -      0      False  2026-09-27 05:5
9:19.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\Sys
tem32\svchost.exe -k LocalSystemNetworkRestricted -p -s NcbService C:\Windows\System32\svchost.
exe
** 2228 864      svchost.exe      0x9e8a6a158080 12     -      0      False  2026-09-27 05:5
9:19.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k LocalServiceNoNetworkFirewall -p      C:\Windows\system32\svchost.exe
** 2868 864      svchost.exe      0x9e8a6f9502c0 5      -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k NetworkService -p -s CryptSvc C:\Windows\system32\svchost.exe
** 2996 864      svchost.exe      0x9e8a6f95e0c0 3      -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\Sys
tem32\svchost.exe -k LocalSystemNetworkRestricted -p -s TrkWks      C:\Windows\System32\svchost.
exe
** 3064 864      svchost.exe      0x9e8a6f9a4080 5      -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k netsvcs -p -s WpnService      C:\Windows\system32\svchost.exe
** 4792 864      svchost.exe      0x9e8a6f9a6080 4      -      0      False  2026-09-27 06:0
4:01.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k netsvcs -p -s TokenBroker      C:\Windows\system32\svchost.exe
** 2876 864      svchost.exe      0x9e8a6f936200 8      -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\Sys
tem32\svchost.exe -k utcsvc -p      C:\Windows\System32\svchost.exe
** 828 864       svchost.exe      0x9e8a6a6432c0 9      -      0      False  2026-09-27 06:0
1:22.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k LocalSystemNetworkRestricted -p -s UALSVC      C:\Windows\system32\svchost.
exe
** 5952 864      svchost.exe      0x9e8a714bd080 8      -      0      False  2026-09-27 06:0
4:12.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k LocalSystemNetworkRestricted -p -s PcaSvc      C:\Windows\system32\svchost.
exe
** 964 864       svchost.exe      0x9e8a6ed49280 7      -      0      False  2026-09-27 05:5
9:19.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k RPCSS -p      C:\Windows\system32\svchost.exe
** 1604 864      svchost.exe      0x9e8a6f572240 2      -      0      False  2026-09-27 05:5
9:19.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k netsvcs -p -s ProfSvc C:\Windows\system32\svchost.exe
** 2884 864      vmtoolsd.exe     0x9e8a6f9a1540 14     -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Program Files\VMware\VMware Tools\vmtoolsd.exe
"C:\Program Files\VMware\VMware Tools\vmtoolsd.exe"      C:\Program Files\VMware\VMware Tools\v
mtoolsd.exe
** 1732 864      svchost.exe      0x9e8a6e326080 3      -      1      False  2026-09-27 06:0
4:01.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k UnistackSvcGroup -s CDPUserSvc      C:\Windows\system32\svchost.exe

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** 6088 864      svchost.exe      0x9e8a6a629340 2      -      0      False  2026-09-27 06:0
4:01.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe  C:\Windows\sys
tem32\svchost.exe -k appmodel -p -s StateRepository      C:\Windows\system32\svchost.exe
** 6984 864      svchost.exe      0x9e8a74752080 1      -      0      False  2026-09-27 06:1
4:19.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe  C:\Windows\Sys
tem32\svchost.exe -k LocalSystemNetworkRestricted -p -s StorSvc      C:\Windows\System32\svchost.
exe
** 1612 864      svchost.exe      0x9e8a6f574240 2      -      0      False  2026-09-27 05:5
9:19.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe  C:\Windows\Sys
tem32\svchost.exe -k netsvcs -p -s Themes      C:\Windows\System32\svchost.exe
** 4684 864      svchost.exe      0x9e8a700612c0 1      -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe  C:\Windows\sys
tem32\svchost.exe -k LocalService -p -s fdPHost      C:\Windows\system32\svchost.exe
** 1872 864      svchost.exe      0x9e8a6f629480 7      -      0      False  2026-09-27 05:5
9:19.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe  C:\Windows\Sys
tem32\svchost.exe -k LocalService -p -s netprofm      C:\Windows\System32\svchost.exe
** 3024 864      svchost.exe      0x9e8a6f96c2c0 5      -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe  C:\Windows\Sys
tem32\svchost.exe -k NetworkService -p -s WinRM      C:\Windows\System32\svchost.exe
** 1620 864      svchost.exe      0x9e8a6f5762c0 4      -      0      False  2026-09-27 05:5
9:19.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe  C:\Windows\sys
tem32\svchost.exe -k LocalService -p -s EventSystem      C:\Windows\system32\svchost.exe
** 1748 864      svchost.exe      0x9e8a6f5bc240 9      -      0      False  2026-09-27 05:5
9:19.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe  C:\Windows\sys
tem32\svchost.exe -k netsvcs -p -s Schedule      C:\Windows\system32\svchost.exe
*** 3240      1748      taskhostw.exe      0x9e8a70b32680 4      -      1      False  2026-09
-27 06:04:01.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\taskhostw.exe  taskho
stw.exe {222A245B-E637-4AE9-A93F-A59CA119A75E}      C:\Windows\system32\taskhostw.exe
*** 6344      1748      MicrosoftEdgeU      0x9e8a71583080 4      -      0      True   2026-09
-27 06:05:02.000000 UTC  N/A      \Device\HarddiskVolume4\Program Files (x86)\Microsoft\EdgeUpda
te\MicrosoftEdgeUpdate.exe "C:\Program Files (x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.ex
e" /c      C:\Program Files (x86)\Microsoft\EdgeUpdate\MicrosoftEdgeUpdate.exe
** 2900 864      svchost.exe      0x9e8a6f99f540 24     -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe  C:\Windows\sys
tem32\svchost.exe -k netsvcs -p -s Winmgmt      C:\Windows\system32\svchost.exe
** 3288 864      svchost.exe      0x9e8a6fa1d240 7      -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe  C:\Windows\Sys
tem32\svchost.exe -k smbvc -s LanmanServer      C:\Windows\System32\svchost.exe
** 2908 864      svchost.exe      0x9e8a6f9a92c0 1      -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe  C:\Windows\sys
tem32\svchost.exe -k LocalService -p -s SstpSvc      C:\Windows\system32\svchost.exe
** 3548 864      svchost.exe      0x9e8a6fb5f300 3      -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe  C:\Windows\sys
tem32\svchost.exe -k NetworkServiceNetworkRestricted -p -s PolicyAgent      C:\Windows\system32
\svchost.exe
** 5084 864      dllhost.exe      0x9e8a7054a240 10     -      0      False  2026-09-27 05:5
9:21.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\dllhost.exe  C:\Windows\sys
tem32\dllhost.exe /Processid:{02D4B3F1-FD88-11D1-960D-00805FC79235}      C:\Windows\system32
\dllhost.exe
** 1504 864      svchost.exe      0x9e8a6f4d42c0 1      -      0      False  2026-09-27 05:5
9:19.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe  C:\Windows\sys
tem32\svchost.exe -k LocalService -p -s nsi      C:\Windows\system32\svchost.exe
** 1760 864      svchost.exe      0x9e8a6f5215c0 2      -      0      False  2026-09-27 05:5
9:19.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe  C:\Windows\sys
tem32\svchost.exe -k netsvcs -p -s SENS      C:\Windows\system32\svchost.exe
** 3040 864      svchost.exe      0x9e8a6f9c12c0 4      -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe  C:\Windows\sys
tem32\svchost.exe -k LocalService -s W32Time      C:\Windows\system32\svchost.exe
** 3168 864      elastic-agent.    0x9e8a6f953080 22     -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Program Files\Elastic\Agent\data\elastic-agent
-9.5.4+build202609161310-b18dae\elastic-agent.exe "C:\Program Files\Elastic\Agent\elastic-agen

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t.exe"      C:\Program Files\Elastic\Agent\elastic-agent.exe
*** 5664      3168      elastic-otel-c  0x9e8a6a06b080  32      -      0      False  2026-09
-27 05:59:22.000000 UTC  N/A      \Device\HarddiskVolume4\Program Files\Elastic\Agent\data\elast
ic-agent-9.5.4+build202609161310-b18dae\components\elastic-otel-collector.exe      "C:\Program
Files\Elastic\Agent\data\elastic-agent-9.5.4+build202609161310-b18dae\components\elastic-otel-c
ollector.exe" --supervised --supervised.monitoring.url=npipe:///ZaV_00Lb1vdHHFhzn0256b16Lra5CAQ
J.sock --feature-gates=telemetry.newPipelineTelemetry --feature-gates=service.partialReload,ser
vice.partialReloadReceivers --supervised.logging.level=info --config=stdingob: --set=extension
s::healthcheckv2/f9144ea3-d2c8-44b4-8c4b-4f4d3cc9f817::http:--endpoint=localhost:49679      C:\P
rogram Files\Elastic\Agent\data\elastic-agent-9.5.4+build202609161310-b18dae\components\elastic
-otel-collector.exe
**** 5508      5664      conhost.exe      0x9e8a70bd2080  4      -      0      False  2026-09
-27 05:59:25.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\conhost.exe      \??
\C:\Windows\system32\conhost.exe 0x4 C:\Windows\system32\conhost.exe
** 2916 864      svchost.exe      0x9e8a6f9a7280  4      -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k netsvcs -p -s IKEEXT C:\Windows\system32\svchost.exe
** 3176 864      Velociraptor.e 0x9e8a6f9a2080  19      -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Program Files\Velociraptor\Velociraptor.exe
"C:\Program Files\Velociraptor\Velociraptor.exe" service run      C:\Program Files\Velociraptor
\Velociraptor.exe
** 5992 864      svchost.exe      0x9e8a70d3f080  2      -      0      False  2026-09-27 06:1
6:23.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\Sys
tem32\svchost.exe -k LocalSystemNetworkRestricted -p -s WdiSystemHost      C:\Windows\System32
\svchost.exe
** 1260 864      svchost.exe      0x9e8a6f70c680  4      -      0      False  2026-09-27 05:5
9:19.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k netsvcs -p -s UserManager      C:\Windows\system32\svchost.exe
*** 1728      1260      sihost.exe      0x9e8a6a14e080  7      -      1      False  2026-09
-27 06:04:01.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\sihost.exe      sihos
t.exe      C:\Windows\system32\sihost.exe
** 2156 864      svchost.exe      0x9e8a6f7142c0  4      -      0      False  2026-09-27 05:5
9:19.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k LocalService -p -s FontCache C:\Windows\system32\svchost.exe
** 2924 864      svchost.exe      0x9e8a6f9a5240  2      -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k LocalSystemNetworkRestricted -p -s SysMain      C:\Windows\system32\svchost.
exe
** 1904 864      svchost.exe      0x9e8a6f65d100  9      -      0      False  2026-09-27 05:5
9:19.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k NetworkService -p -s Dnscache C:\Windows\system32\svchost.exe
** 5104 864      svchost.exe      0x9e8a70549080  5      -      0      False  2026-09-27 05:5
9:21.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k LocalServiceAndNoImpersonation -p -s FDResPub C:\Windows\system32\svchost.
exe
** 2932 864      wlms.exe      0x9e8a6f9ab240  2      -      0      False  2026-09-27 05:5
9:20.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\wlms\wlms.exe      C:\Windows\sys
tem32\wlms\wlms.exeC:\Windows\system32\wlms\wlms.exe
** 1912 864      svchost.exe      0x9e8a6f660100  4      -      0      False  2026-09-27 05:5
9:19.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k LocalServiceNetworkRestricted -p      C:\Windows\system32\svchost.exe
** 124 864      svchost.exe      0x9e8a6d841080  9      -      0      False  2026-09-27 05:5
9:18.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\svchost.exe      C:\Windows\sys
tem32\svchost.exe -k DcomLaunch -p C:\Windows\system32\svchost.exe
*** 3972      124      ShellExperienc 0x9e8a70e4a240  30      -      1      False  2026-09
-27 06:04:03.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\SystemApps\ShellExperienceHost
_cw5n1h2txyewy\ShellExperienceHost.exe      "C:\Windows\SystemApps\ShellExperienceHost_cw5n1h2tx
yewy\ShellExperienceHost.exe" -ServerName:App.AppXtk181tbxbce2qsex02s8tw7hfxa9xb3t.mca
C:\Windows\SystemApps\ShellExperienceHost_cw5n1h2txyewy\ShellExperienceHost.exe
*** 4584      124      unsecapp.exe      0x9e8a7001d200  1      -      0      False  2026-09
-27 05:59:20.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\wbem\unsecapp.exe

```

```

C:\Windows\system32\wbem\unsecapp.exe -Embedding C:\Windows\system32\wbem\unsecapp.exe
*** 5480      124      RuntimeBroker.  0x9e8a71477080  1      -      1      False  2026-09
-27 06:04:10.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\RuntimeBroker.exe
C:\Windows\System32\RuntimeBroker.exe -Embedding C:\Windows\System32\RuntimeBroker.exe
*** 4008      124      smartscreen.ex  0x9e8a70bd0080  8      -      1      False  2026-09
-27 06:04:12.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\smartscreen.exe
C:\Windows\System32\smartscreen.exe -Embedding C:\Windows\System32\smartscreen.exe
*** 4824      124      WmiPrvSE.exe   0x9e8a70109240  10     -      0      False  2026-09
-27 05:59:20.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\wbem\WmiPrvSE.exe
C:\Windows\system32\wbem\wmiprvse.exe C:\Windows\system32\wbem\wmiprvse.exe
*** 2768      124      RuntimeBroker.  0x9e8a70f42300  2      -      1      False  2026-09
-27 06:04:03.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\RuntimeBroker.exe
C:\Windows\System32\RuntimeBroker.exe -Embedding C:\Windows\System32\RuntimeBroker.exe
*** 1844      124      RuntimeBroker.  0x9e8a70e94300  1      -      1      False  2026-09
-27 06:04:04.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\RuntimeBroker.exe
C:\Windows\System32\RuntimeBroker.exe -Embedding C:\Windows\System32\RuntimeBroker.exe
*** 5144      124      dllhost.exe    0x9e8a705e9280  4      -      0      False  2026-09
-27 05:59:21.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\System32\dllhost.exe C:\Win
dows\system32\DllHost.exe /Processid:{3EB3C877-1F16-487C-9050-104DBCD66683} C:\Windows\s
ystem32\DllHost.exe
*** 1144      124      SearchUI.exe   0x9e8a6ebcf0c0  17     -      1      False  2026-09
-27 06:04:03.000000 UTC  N/A      \Device\HarddiskVolume4\Windows\SystemApps\Microsoft.Windows.C
ortana_cw5nlh2txyewy\SearchUI.exe "C:\Windows\SystemApps\Microsoft.Windows.Cortana_cw5nlh2txy
ewy\SearchUI.exe" -ServerName:CortanaUI.AppXa50dqqag5gqv4a428c9y1jjw7m3btvepj.mca C:\Windows
\SystemApps\Microsoft.Windows.Cortana_cw5nlh2txyewy\SearchUI.exe
* 900 720 lsass.exe 0x9e8a6de130c0 7 - 0 False 2026-09-27 05:5
9:18.000000 UTC N/A \Device\HarddiskVolume4\Windows\System32\lsass.exe C:\Windows\sys
tem32\lsass.exe C:\Windows\system32\lsass.exe
* 884 720 WerFault.exe 0x9e8a6de020c0 0 - 0 False 2026-09-27 05:5
9:18.000000 UTC 2026-09-27 05:59:28.000000 UTC \Device\HarddiskVolume4\Windows\System32\WerFa
ult.exe - -
* 108 720 fontdrvhost.ex 0x9e8a6ec38140 5 - 0 False 2026-09-27 05:5
9:18.000000 UTC N/A \Device\HarddiskVolume4\Windows\System32\fontdrvhost.exe "fontd
rvhost.exe" C:\Windows\system32\fontdrvhost.exe
740 712 csrss.exe 0x9e8a6ec0c140 11 - 1 False 2026-09-27 05:5
9:18.000000 UTC N/A \Device\HarddiskVolume4\Windows\System32\csrss.exe %SystemRoot%\s
ystem32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,20480,768 Windows=On SubSystemTyp
e=Windows ServerDll=basesrv,1 ServerDll=winsrv:UserServerDllInitialization,3 ServerDll=sxssrv,4
ProfileControl=Off MaxRequestThreads=16 C:\Windows\system32\csrss.exe
820 712 winlogon.exe 0x9e8a6ec2e100 3 - 1 False 2026-09-27 05:5
9:18.000000 UTC N/A \Device\HarddiskVolume4\Windows\System32\winlogon.exe winlogon.exe
C:\Windows\system32\winlogon.exe
* 5568 820 userinit.exe 0x9e8a7003e080 0 - 1 False 2026-09-27 06:0
4:02.000000 UTC 2026-09-27 06:04:23.000000 UTC \Device\HarddiskVolume4\Windows\System32\useri
nit.exe - -
** 5516 5568 explorer.exe 0x9e8a708c2080 32 - 1 False 2026-09-27 06:0
4:02.000000 UTC N/A \Device\HarddiskVolume4\Windows\explorer.exe C:\Windows\Explorer.EX
E C:\Windows\Explorer.EXE
*** 1448 5516 msedge.exe 0x9e8a714b6400 0 - 1 False 2026-09
-27 06:04:12.000000 UTC 2026-09-27 06:04:18.000000 UTC \Device\HarddiskVolume4\Program Files
(x86)\Microsoft\Edge\Application\msedge.exe - -
*** 6188 5516 vmtoolsd.exe 0x9e8a71581080 6 - 1 False 2026-09
-27 06:04:14.000000 UTC N/A \Device\HarddiskVolume4\Program Files\VMware\VMware Tools\vmto
olsd.exe "C:\Program Files\VMware\VMware Tools\vmtoolsd.exe" -n vmusr C:\Program Files\VMw
are\VMware Tools\vmtoolsd.exe
*** 6320 5516 powershell.exe 0x9e8a7474a080 10 - 1 False 2026-09
-27 06:04:45.000000 UTC N/A \Device\HarddiskVolume4\Windows\System32\WindowsPowerShell\v1.
0\powershell.exe "C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe" C:\Windows\S
ystem32\WindowsPowerShell\v1.0\powershell.exe
**** 6268 6320 conhost.exe 0x9e8a74749080 3 - 1 False 2026-09
-27 06:04:45.000000 UTC N/A \Device\HarddiskVolume4\Windows\System32\conhost.exe \??

```

```

\Device\HarddiskVolume4\Velociraptor\velociraptor.exe "C:\Velociraptor\velociraptor.exe" --config .\server.config.yaml frontend -v C:\Velociraptor\velociraptor.exe
**** 6660 6320 velociraptor.e 0x9e8a74754080 11 - 1 False 2026-09-27 06:04:48.000000 UTC N/A
**** 6944 6660 velociraptor.e 0x9e8a74753080 19 - 1 False 2026-09-27 06:04:49.000000 UTC N/A
*** 6236 5516 a3.exe 0x9e8a7474f080 19 - 1 False 2026-09-27 06:33:08.000000 UTC N/A
**** 7096 6236 conhost.exe 0x9e8a6f954080 4 - 1 False 2026-09-27 06:33:08.000000 UTC N/A
**** 1156 6236 cmd.exe 0x9e8a70cb2080 0 - 1 False 2026-09-27 06:35:45.000000 UTC 2026-09-27 06:35:45.000000 UTC
**** 3408 1156 ipconfig.exe 0x9e8a6fd0c640 0 - 1 False 2026-09-27 06:35:45.000000 UTC 2026-09-27 06:35:45.000000 UTC
**** 1200 6236 cmd.exe 0x9e8a70c86680 0 - 1 False 2026-09-27 06:33:48.000000 UTC 2026-09-27 06:33:48.000000 UTC
**** 6872 1200 ipconfig.exe 0x9e8a6ec0a140 0 - 1 False 2026-09-27 06:33:48.000000 UTC 2026-09-27 06:33:48.000000 UTC
* 148 820 fontdrvhost.ex 0x9e8a6d84c080 5 - 1 False 2026-09-27 05:59:18.000000 UTC N/A
* 1100 820 dwm.exe 0x9e8a6eda0080 16 - 1 False 2026-09-27 05:59:19.000000 UTC N/A

```

A4 — Volatility windows.cmdline --pid 6236

```

PS C:\Users\PC\Documents\Voltality_analysting\uploads\auto> vol -f .\PhysicalMemory.dd windows.cmdline --pid 6236
Volatility 3 Framework 2.28.0
Progress: 100.00 PDB scanning finished
PID Process Args
6236 a3.exe "C:\Users\Administrator\Desktop\a3.exe"

```

A5 — Volatility windows.dlllist --pid 6236 — full output

```

PS C:\Users\PC\Documents\Voltality_analysting\uploads\auto> vol -f .\PhysicalMemory.dd windows.
dlllist --pid 6236
Volatility 3 Framework 2.28.0
Progress: 100.00 PDB scanning finished
PID Process Base Size Name Path LoadCount LoadTime File output
6236 a3.exe 0x142366d0000 0x1aa000 a3.exe C:\Users\Administrator\Desktop\a3.exe
-1 2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff423c0000 0x1ee000 ntdll.dll C:\Windows\SYSTEM32\ntdll.dll
-1 2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff3cda0000 0x64000 MSCOREE.DLL C:\Windows\SYSTEM32\MSCOREE.DLL -1
2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff41c80000 0xb3000 KERNEL32.dll C:\Windows\System32\KERNEL32.dll
-1 2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff3e5a0000 0x29d000 KERNELBASE.dll C:\Windows\System32\KERNELBASE.
dll -1 2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff41330000 0xa7000 ADVAPI32.dll C:\Windows\System32\ADVAPI32.dll
-1 2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff410c0000 0x9e000 msvcrt.dll C:\Windows\System32\msvcrt.dll -1
2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff41770000 0x9f000 sechost.dll C:\Windows\System32\sechost.dll -1
2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff42270000 0x11d000 RPCRT4.dll C:\Windows\System32\RPCRT4.dll
-1 2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff3cc40000 0x9c000 mscoreei.dll C:\Windows\Microsoft.NET\Framework64\v
4.0.30319\mscoreei.dll 6 2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff416b0000 0x52000 SHLWAPI.dll C:\Windows\System32\SHLWAPI.dll -1
2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff41950000 0x32b000 combase.dll C:\Windows\System32\combase.dll
-1 2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff3e450000 0xfa000 ucrtbase.dll C:\Windows\System32\ucrtbase.dll
-1 2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff3f510000 0x82000 bcryptPrimitives.dll C:\Windows\System32\bcryptPrimi
tives.dll -1 2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff40d70000 0x29000 GDI32.dll C:\Windows\System32\GDI32.dll -1
2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff3f020000 0x1a0000 gdi32full.dll C:\Windows\System32\gdi32full.d
ll 6 2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff3f1c0000 0xa0000 msvcp_win.dll C:\Windows\System32\msvcp_win.dll
-1 2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff413e0000 0x198000 USER32.dll C:\Windows\System32\USER32.dll
6 2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff3f000000 0x20000 win32u.dll C:\Windows\System32\win32u.dll -1
2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff3f6d0000 0x2e000 IMM32.DLL C:\Windows\System32\IMM32.DLL 6
2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff3e400000 0x11000 kernel.appcore.dll C:\Windows\System32\kernel.appc
ore.dll 6 2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff3c490000 0xa000 VERSION.dll C:\Windows\SYSTEM32\VERSION.dll 6
2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff2a0a0000 0xa3b000 clr.dll C:\Windows\Microsoft.NET\Framework64\v
4.0.30319\clr.dll -1 2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff3cac0000 0xf7000 MSVCR120_CLR0400.dll C:\Windows\SYSTEM32\MSVCR120_CL
R0400.dll -1 2026-09-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff28b00000 0x1591000 mscorlib.ni.dll C:\Windows\assembly\NativeImage
s_v4.0.30319_64\mscorlib\bee85cfb2459dd2aeb6bbd03e279c875\mscorlib.ni.dll 6 2026-0
9-27 06:33:08.000000 UTC Disabled
6236 a3.exe 0x7fff41160000 0x156000 ole32.dll C:\Windows\System32\ole32.dll

```

```

6      2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff280b0000 0x12b000 clrjit.dll C:\Windows\Microsoft.NET\Framework
ork64\v4.0.30319\clrjit.dll 6      2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff27260000 0xc44000 System.ni.dll C:\Windows\assembly\NativeImage
s_v4.0.30319_64\System\369c0041b94fe8a77a049e07a31504d2\System.ni.dll 6      2026-09-27 06:
33:08.000000 UTC Disabled
6236  a3.exe 0x7fff3d9f0000 0x29000 wldp.dll C:\Windows\SYSTEM32\wldp.dll 6
2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff3f310000 0x1f5000 CRYPT32.dll C:\Windows\System32\CRYPT32.dll
6      2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff3e380000 0x12000 MSASN1.dll C:\Windows\System32\MSASN1.dll 6
2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff3ef90000 0x61000 WINTRUST.dll C:\Windows\System32\WINTRUST.dll
6      2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff3fd00000 0x1b000 CRYPTSP.dll C:\Windows\System32\CRYPTSP.dll 6
2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff3d310000 0x33000 rsaenh.dll C:\Windows\system32\rsaenh.dll 6
2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff3f5a0000 0x26000 bcrypt.dll C:\Windows\System32\bcrypt.dll -1
2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff3d960000 0xc000 CRYPTBASE.dll C:\Windows\SYSTEM32\CRYPTBASE.dll
6      2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff41580000 0xa9000 clbcatq.dll C:\Windows\System32\clbcatq.dll 6
2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff3f870000 0x14f8000 shell32.dll C:\Windows\System32\shell32.dll
6      2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff3e550000 0x4a000 cfgmgr32.dll C:\Windows\System32\cfgmgr32.dll
6      2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff41d40000 0xa7000 shcore.dll C:\Windows\System32\shcore.dll 6
2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff3e840000 0x749000 windows.storage.dll C:\Windows\System32\win
dows.storage.dll 6      2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff3e420000 0x23000 profapi.dll C:\Windows\System32\profapi.dll 6
2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff3e3a0000 0x5d000 powrprof.dll C:\Windows\System32\powrprof.dll
6      2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff24550000 0xa50000 System.Core.ni.dll C:\Windows\assembly\Nat
iveImages_v4.0.30319_64\System.Core\4d51711570b44dabfc66beb6da8057d8\System.Core.ni.dll 6
2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff3e280000 0x29000 userenv.dll C:\Windows\SYSTEM32\userenv.dll 6
2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff3d1e0000 0xc000 Secur32.dll C:\Windows\SYSTEM32\Secur32.dll 6
2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff3e2b0000 0x2f000 SSPICLI.DLL C:\Windows\SYSTEM32\SSPICLI.DLL 6
2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff25a30000 0x12b000 System.Configuration.ni.dll C:\Windows\asse
mbly\NativeImages_v4.0.30319_64\System.Configuration\0de5a4ecad78673b41ceafe884cde7e8\System.Co
nfiguration.ni.dll 6      2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff22480000 0x88c000 System.Xml.ni.dll C:\Windows\assembly\Nat
iveImages_v4.0.30319_64\System.Xml\2928f97eb11f536b35c532690cdb9f46\System.Xml.ni.dll 6
2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff41810000 0x8000 psapi.dll C:\Windows\System32\psapi.dll 6
2026-09-27 06:33:08.000000 UTC Disabled
6236  a3.exe 0x7fff412c0000 0x6d000 ws2_32.dll C:\Windows\System32\ws2_32.dll -1
2026-09-27 06:33:09.000000 UTC Disabled
6236  a3.exe 0x7fff3d790000 0x67000 mswsock.dll C:\Windows\system32\mswsock.dll 6
2026-09-27 06:33:09.000000 UTC Disabled
6236  a3.exe 0x7fff3cd60000 0x16000 napinsp.dll C:\Windows\system32\napinsp.dll 6
2026-09-27 06:33:09.000000 UTC Disabled
6236  a3.exe 0x7fff3d4e0000 0xc6000 DNSAPI.dll C:\Windows\SYSTEM32\DNSAPI.dll -1
2026-09-27 06:33:09.000000 UTC Disabled

```

```

6236 a3.exe 0x7fff40da0000 0x8000 NSI.dll C:\Windows\System32\NSI.dll -1 2026-09
-27 06:33:09.000000 UTC Disabled
6236 a3.exe 0x7fff3d4a0000 0x3d000 IPHLPAPI.DLL C:\Windows\SYSTEM32\IPHLPAPI.DLL
-1 2026-09-27 06:33:09.000000 UTC Disabled
6236 a3.exe 0x7fff3ca80000 0xe000 winnr.dll C:\Windows\System32\winnr.dll 6
2026-09-27 06:33:09.000000 UTC Disabled
6236 a3.exe 0x7fff38cf0000 0x1b000 NLAapi.dll C:\Windows\system32\NLAapi.dll 6
2026-09-27 06:33:09.000000 UTC Disabled
6236 a3.exe 0x7fff3a730000 0x15000 wshbth.dll C:\Windows\system32\wshbth.dll 6
2026-09-27 06:33:09.000000 UTC Disabled
6236 a3.exe 0x7fff36d10000 0x16000 dhcpcsvc6.DLL C:\Windows\SYSTEM32\dhcpcsvc6.DLL
-1 2026-09-27 06:33:09.000000 UTC Disabled
6236 a3.exe 0x7fff36ea0000 0x1c000 dhcpcsvc.DLL C:\Windows\SYSTEM32\dhcpcsvc.DLL
-1 2026-09-27 06:33:09.000000 UTC Disabled
6236 a3.exe 0x7fff37af0000 0xb000 WINNSI.DLL C:\Windows\SYSTEM32\WINNSI.DLL -1
2026-09-27 06:33:09.000000 UTC Disabled
6236 a3.exe 0x7fff2d210000 0x333000 System.Runtime.Serialization.ni.dll C:\Wind
ows\assembly\NativeImages_v4.0.30319_64\System.Runteb92aa12#\4b8506ace4a656a2119cf57c38a8ab15\S
ystem.Runtime.Serialization.ni.dll 6 2026-09-27 06:33:09.000000 UTC Disabled
6236 a3.exe 0x7fff3a700000 0x27000 SMDiagnostics.ni.dll C:\Windows\assembly\NativeImage
s_v4.0.30319_64\SMDiagnostics\aa02dc6f7bb0c92feecb04dfa7d898d8\SMDiagnostics.ni.dll 6
2026-09-27 06:33:09.000000 UTC Disabled
6236 a3.exe 0x7fff38aa0000 0xf4000 System.ServiceModel.Internals.ni.dll C:\Windows\asse
mbly\NativeImages_v4.0.30319_64\System.Servd1dec626#\15398b2b53bb1e79306881986af0bcbe\System.Se
rviceModel.Internals.ni.dll 6 2026-09-27 06:33:09.000000 UTC Disabled
6236 a3.exe 0x7fff2e960000 0xf1000 rasapi32.dll C:\Windows\SYSTEM32\rasapi32.dll
6 2026-09-27 06:33:09.000000 UTC Disabled
6236 a3.exe 0x7fff2eb80000 0x33000 rasman.dll C:\Windows\SYSTEM32\rasman.dll 6
2026-09-27 06:33:09.000000 UTC Disabled
6236 a3.exe 0x7fff31280000 0x15000 rtutils.dll C:\Windows\SYSTEM32\rtutils.dll 6
2026-09-27 06:33:09.000000 UTC Disabled
6236 a3.exe 0x7fff35d60000 0xff000 winhttp.dll C:\Windows\SYSTEM32\winhttp.dll 6
2026-09-27 06:33:09.000000 UTC Disabled
6236 a3.exe 0x7fff41820000 0xc4000 oleaut32.dll C:\Windows\System32\oleaut32.dll
6 2026-09-27 06:33:09.000000 UTC Disabled
6236 a3.exe 0x7fff3d240000 0x8c000 schannel.DLL C:\Windows\system32\schannel.DLL
6 2026-09-27 06:33:09.000000 UTC Disabled
6236 a3.exe 0x7fff34c10000 0x15000 mskeyprotect.dll C:\Windows\SYSTEM32\mskeyprotec
t.dll 6 2026-09-27 06:33:09.000000 UTC Disabled
6236 a3.exe 0x7fff3da60000 0x2c000 ncrypt.dll C:\Windows\SYSTEM32\ncrypt.dll 6
2026-09-27 06:33:09.000000 UTC Disabled
6236 a3.exe 0x7fff3da20000 0x3c000 NTASN1.dll C:\Windows\SYSTEM32\NTASN1.dll 6
2026-09-27 06:33:09.000000 UTC Disabled
6236 a3.exe 0x7fff34cc0000 0x24000 ncryptsslp.dll C:\Windows\system32\ncryptsslp.dll
6 2026-09-27 06:33:09.000000 UTC Disabled
6236 a3.exe 0x7fff3c4a0000 0x22000 gpapi.dll C:\Windows\SYSTEM32\gpapi.dll 6
2026-09-27 06:33:09.000000 UTC Disabled

```

A6 — Volatility windows.filescan — matches for a3.exe

```

PS C:\Users\PC\Documents\Voltality_analysting\uploads\auto> Select-String -Path .\filescan.txt
-Pattern "a3\.exe"
filesan.txt:2076:0x9e8a705baeb0 \Users\Administrator\Desktop\a3.exe
filesan.txt:2091:0x9e8a705bcad0 \Users\Administrator\Desktop\a3.exe
filesan.txt:2094:0x9e8a705bd110 \Users\Administrator\Desktop\a3.exe

```

A7 — Volatility windows.malfind --pid 6236 — region table

```
vol -q -f .\PhysicalMemory.dd windows.malfind --pid 6236
Volatility 3 Framework 2.28.0
```

PID	Process	Start VPN	End VPN	Tag	Protection	CommitCharge	PrivateMemory	File output
6236	a3.exe	0x142386b0000	0x142386bffff	VadS	PAGE_EXECUTE_READWRITE	2	1	Disabled
6236	a3.exe	0x14238680000	0x1423868ffff	VadS	PAGE_EXECUTE_READWRITE	2	1	Disabled
6236	a3.exe	0x7ff4f1550000	0x7ff4f15effff	VadS	PAGE_EXECUTE_READWRITE	2	1	Disabled
6236	a3.exe	0x7ff4f1540000	0x7ff4f154ffff	VadS	PAGE_EXECUTE_READWRITE	1	1	Disabled

[Hexdump and disassembly columns are retained in the original analyst log and are not reproduced in this report.]

A8 — Sysmon Event ID 3 — transcription of Figure 4

Log Name: Microsoft-Windows-Sysmon/Operational
Source: Sysmon
Event ID: 3
Task Category: Network connection detected (rule: NetworkConnect)
Computer: WIN-50SDUJBR958
Logged: 27/09/2026 1:17:32 PM (UTC+07:00)
Filter: Event ID 3 -> 1,068 events in log

Network connection detected:
RuleName: technique_id=T1036,technique_name=Masquerading
UtcTime: 2026-09-27 06:17:31.247
ProcessGuid: {388cdf0-b4fa-6ab8-dd00-00000000d00}
ProcessId: 832
Image: C:\Users\Administrator\Desktop\a3.exe
User: WIN-50SDUJBR958\Administrator
Protocol: tcp
Initiated: true
SourceIsIpv6: false
SourceIp: 10.10.12.20
SourcePort: 49967
DestinationIsIpv6: false
DestinationIp: 192.168.1.7
DestinationPort: 443

A9 — Trigger script run — transcription of Figure 2

```
(venv) root@ubuntuserver:~/soc-lab# python3 suricata_velociraptor_trigger.py --once --lookback 600
CONNECTION CHECK
Elasticsearch OK - 3,206 alerts in 'suricata-eve*'
Velociraptor API OK - 10.10.12.20:8001 (server host: WIN-50SDUJBR958)
Victim 10.10.12.20 -> C.8f3be93e17030665 (WIN-50SDUJBR958)

SURICATA -> VELOCIRAPTOR TRIGGER STARTED
ES:      http://10.10.12.11:9200/suricata-eve*
Victim:   10.10.12.20
Artifact: Windows.Memory.Acquisition
Poll:     15s | Cooldown: 30 min
Mode:     LIVE / ONE-SHOT

--- Cycle #1 | 06:39:46 ---
[sev 1] C2 Mythic TLS Cert | 192.168.1.7:443 -> 10.10.12.20:50083
TRIGGERED Windows.Memory.Acquisition | INC-20260927-063946 | client C.8f3be93e17030665 | flow F.DASBKCL6L74KI

Cycles: 1 | Alerts: 1 | Dumps triggered: 1 | Finished: 0 | Failed/Timeout: 0
1 flow still running on Velociraptor - see GUI:
INC-20260927-063946 -> flow F.DASBKCL6L74KI
```

A10 — Velociraptor flows and acquisition metadata — transcription of Figure 3

Client: WIN-50SDUJBR958 (C.8f3be93e17030665) - Connected

State	FlowId	Artifact	Created	Last Active
done	F.DASC87A9HHH5K	Windows.Memory.Acquisition	2026-09-27T07:22:05.041Z	2026-09-27T07:23:37.546Z
	nguyen	5120 Mb		
done	F.DASBKCL6L74KI	Windows.Memory.Acquisition	2026-09-27T06:39:46.203Z	2026-09-27T06:41:41.703Z
	nguyen	5120 Mb		
done	F.DAP2N54H5TU9G	Windows.System.PowerShell	2026-09-22T07:17:40.332Z	2026-09-22T07:17:40.853Z
	nguyen	7 Kb		
done	F.DAP025N7KJ4JU	Generic.Client.Info	2026-09-22T04:16:22.315Z	2026-09-22T04:16:23.879Z
	InterrogationService	0 b		

Flow F.DASBKCL6L74KI (analysed image):

Create Time	2026-09-27T06:39:46.203Z
Start Time	2026-09-27T06:39:46.244Z
Last Active	2026-09-27T06:41:41.703Z
Duration	115.46 seconds
State	Completed
Timeout	600 seconds
Uploaded Bytes	5368709120 / 5368709120
Files uploaded	1
Download	WIN-50SDUJBR958-C.8f3be93e17030665-F.DASBKCL6L74KI
Uncompressed	5120 Mb
Compressed	1415 Mb
Container files	8
Started	2026-09-27T06:41:47Z
Duration (s)	90
SHA256	d9d669078ee6df53f287d2dcac100556516a65b75111ca7ab6c8579be48e432c

A11 — AI-SIEM detection output — transcription of Figure 5

```
--- Cycle #10 | 2026-09-28 08:54:37 | Since: 2026-09-28T08:53:53.402Z ---
Pulled 4 new log entries
SAFE: 2 LOW: 2 MEDIUM: 0 HIGH: 0 CRITICAL: 0

Threat #1 [LOW] - Score: 38.5/100 (2 events: 2 LOW)
File:      a3.exe
Path:      C:\Users\Administrator\Desktop\a3.exe
Extension: exe
Size:      0 bytes
SHA256:    4d0ad28aea96e37a3a548c210bb7fb049ddd22a1d2aa61954ba2696f9a877b48
MD5:      4cc8c5e06240f380709b690ff0de8f4b
Agent:     WIN-50SDUJBR958
Time:      2026-09-28T08:53:58.658Z -> 2026-09-28T08:53:58.679Z
Threat:    Suspicious Executable

Detection engines:
[ML Engine] Random Forest: 77.0% probability of malware
Active features (5): is_executable=1 (24.6%), path_depth=4 (2.7%),
                    file_size=0B (2.1%), is_small=1 (0.5%), is_zero_size=1 (0.0%)
+22 features = 0 (70.1% of remaining importance)
[Size Engine] Executable file has size 0 bytes; may be a placeholder or a file being written

Risk breakdown: ML=77.0x50% + IOC=0.0x20% + MITRE=0.0x30% = 38.5
Classification: 38.5 -> LOW (SAFE<20 | LOW 20-39 | MEDIUM 40-59 | HIGH 60-79 | CRITICAL>=80)

Found 1 threat(s) (2 events). Sent 1 deduplicated alert to Shuffle: a3.exe | LOW | 38.5 | WIN-50SDUJBR958
```

A12 — Wireshark TCP conversations — transcription of Figure 6

```
Wireshark > Statistics > Conversations > TCP (E2, traffic.pcap) - top rows
Address A      Port A  Address B      Port B  Packets  Bytes  Pkts A->B  Bytes A->B  Pkts B->A
Bytes B->A  Rel Start  Duration
10.10.12.20    50447    192.168.1.7    443     889     161 kB  381        83 kB      508
78 kB        1.284410   379.8882
10.10.12.11    37222    10.10.12.20    8001    131     15 kB   67         7 kB      64
8 kB         0.000000   367.9454
10.10.12.1     64575    10.10.12.20    8889    68      36 kB   24         5 kB      44
31 kB        113.467043 15.0328
10.10.12.20    49678    10.10.12.11    8220    55      6 kB    28         4 kB      27
2 kB         13.838638 362.9943
10.10.12.20    50457    10.10.12.11    9200    55      62 kB   46        61 kB      9
848 bytes    33.978231 3.1463
(... remaining 77 conversations: 10.10.12.20 <-> 10.10.12.11:9200 Elastic Agent,
10.10.12.1 <-> 10.10.12.20:8889 Velociraptor GUI - lab management traffic)
Totals: Ethernet 4, IPv4 6, TCP 82, UDP 2
```

A13 — C2 console — lab ground truth, transcription of Figure 7

```
LAB GROUND TRUTH - C2 operator console (not investigative evidence)
Callbacks 1-9: IP 10.10.12.20 | Host WIN-50SDUJBR958 | User Administrator | Domain WIN-50SDUJBR
958
Task (callback C-9, agent apollo, operator mythic_admin): shell ipconfig/all
Output: Ethernet0 - Intel(R) 82574L, MAC 00-0C-29-8C-6A-95, DHCP No,
        IPv4 10.10.12.20, Mask 255.255.255.0, Gateway 10.10.12.2, DNS 10.10.12.2
```

A14 — Evidence directory listing and hashes — Figure 1

```
/root/soc-lab/evidence/INC-20260927-072205/
Name           Size      Changed (UTC+07:00)    Rights    Owner
alert.json     1 KB      9/27/2026 2:22:05 PM  rw-r--r-- root
traffic.pcap   1,702 KB  9/27/2026 2:30:19 PM  rw-r--r-- root
traffic.pcap.log 1 KB      9/27/2026 2:22:05 PM  rw-r--r-- root

SHA256 (taken on analyst workstation, 2026-09-28):
alert.json     BD49E274D8DA4133A1EF73A7448079FDEB88C098A155B280D2A9918CBBCF7E89
traffic.pcap   7FA29AC565E48308D4059391B0D4F46DE552B6FF4971DE41BDD93618E4B90879
```

References - NIST SP 800-86 — Guide to Integrating Forensic Techniques into Incident Response. - NIST SP 800-61 Rev. 3 — Incident Response Recommendations and Considerations for Cybersecurity Risk Management. - ISO/IEC 27037:2012 — Identification, collection, acquisition and preservation of digital evidence. - ISO/IEC 27041:2015 — Guidance on assuring suitability and adequacy of investigation methods. - ISO/IEC 27042:2015 — Guidelines for the analysis and interpretation of digital evidence. - MITRE ATT&CK — <https://attack.mitre.org/> - Volatility 3 — <https://volatility3.readthedocs.io/>